

TICVAI

Recommendation & Personalization

Dashboard requirements and functional specification

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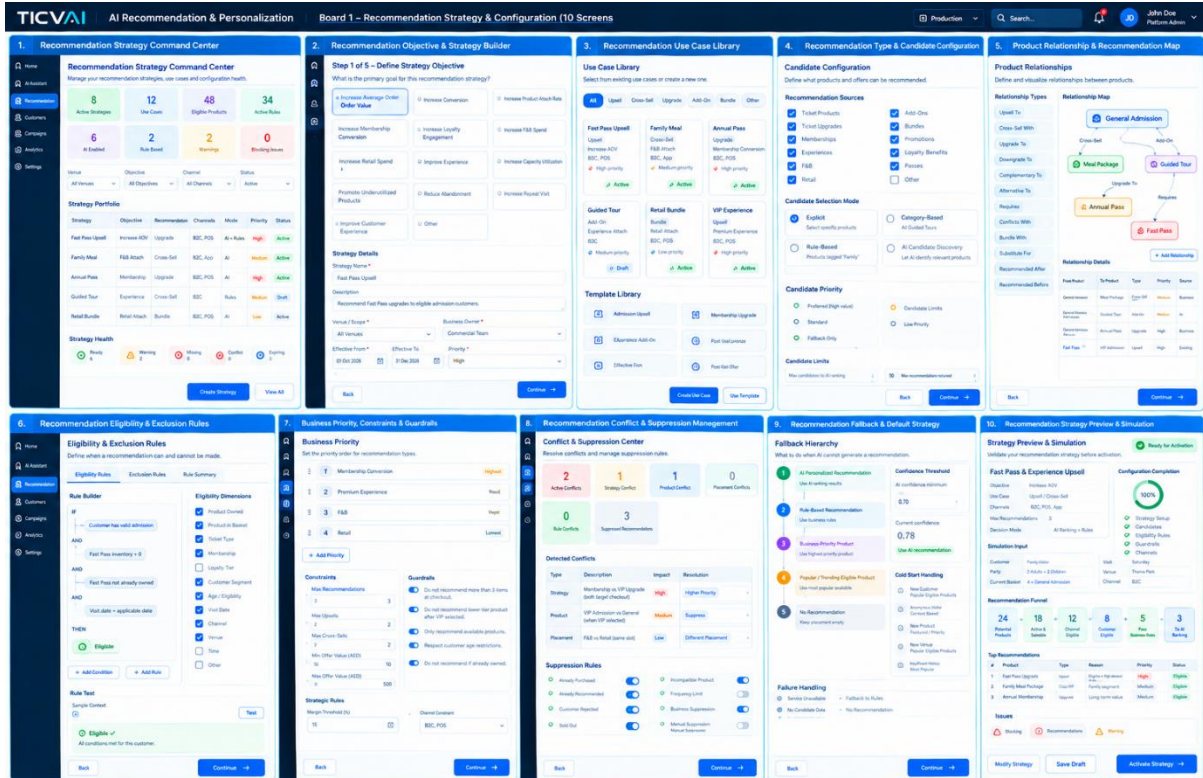
Recommendation & Personalization — Board 1: Recommendation Strategy & Configuration

Module: AI Recommendation & Personalization

Board: 1 of 5

Screens: 10

Board Type: Recommendation Strategy / Use Cases / Eligibility / Business Rules / Product Relationships / Governance



Board Purpose

Board 1 establishes the **business strategy and configuration foundation** for TICVAI's Recommendation & Personalization Engine.

Before AI decides **which customer should receive which recommendation**, TICVAI first needs to understand:

What are we trying to achieve? What are we allowed to recommend? Under what business conditions? What should never be recommended? What takes priority? What should happen when AI cannot make a confident recommendation?

For example, a theme park could configure:

Objective: Increase average transaction value

Recommendation: Fast Pass Upgrade

Audience: Eligible admission customers

Timing: Cart / Checkout / POS

Eligibility: Valid same-day admission

Exclusion: Fast Pass already owned

Capacity Condition: Fast Pass inventory available

Priority: High

Fallback: Meal Package

AI: Allowed to rank eligible recommendations

This board defines the **strategy and guardrails**.

It does **not yet decide the recommendation for a specific customer**. Customer/context intelligence belongs to Board 2 and AI decisioning/ranking belongs to Board 3.

Screen 1 — Recommendation Strategy Command Center

Purpose

Provide management with one centralized view of all recommendation strategies, business objectives, use cases, active policies, coverage, and configuration health.

Header KPIs

- Active Strategies
- Recommendation Use Cases
- Eligible Products / Offers
- Active Rules
- AI-Enabled Strategies
- Rule-Based Strategies
- Configuration Warnings
- Coverage Gaps

Filters

- Venue
- Business Unit
- Recommendation Objective
- Recommendation Type
- Channel
- Journey Stage
- Strategy Status
- AI / Rule Based
- Effective Date

Strategy Portfolio

Example:

Strategy	Objective	Recommendation	Channels	Decision Mode	Priority	Status
Fast Pass Upsell	Increase AOV	Upgrade	B2C/POS	AI + Rules	High	Active
Family Meal	F&B Attach	Cross-Sell	B2C/App	AI	Medium	Active

Strategy	Objective	Recommendation	Channels	Decision Mode	Priority	Status
Annual Pass	Membership Conversion	Upgrade	B2C/POS	AI	High	Active
Guided Tour	Experience Attach	Cross-Sell	B2C	Rules	Medium	Draft

Strategy Health

Each strategy should show:

- Configuration Complete
- Missing Eligibility
- Missing Placement
- No Eligible Products
- Conflict Detected
- Expiring
- Ready

Objective Distribution

Visualize strategies by:

Revenue

Conversion

Experience

Retention

Utilization

Membership

Quick Actions

Create Strategy | Create Use Case | Manage Eligibility | View Conflicts | Simulate Strategy

Screen 2 — Recommendation Objective & Strategy Builder

Purpose

Define **why** TICVAI should generate a recommendation and establish the business objective behind it.

Strategy Wizard

Step 1 — Objective

Select:

- Increase Average Order Value
- Increase Conversion

- Increase Product Attach Rate
- Increase Membership Conversion
- Increase Loyalty Engagement
- Increase F&B Spend
- Increase Retail Spend
- Increase Experience / Attraction Usage
- Improve Capacity Utilization
- Promote Underutilized Products
- Improve Customer Experience
- Reduce Abandonment
- Increase Repeat Visit
- Other Configurable Objective

Strategy Information

- Strategy Name
- Description
- Venue / Scope
- Business Owner
- Effective From
- Effective To
- Priority
- Status

Example

Strategy

Fast Pass Upgrade

Objective

Increase Average Order Value

Secondary Objective

Improve Guest Experience

Success Intent

Define desired business result:

Primary KPI

Upgrade Conversion Rate

Secondary:

- Incremental Revenue
- Attach Rate
- AOV Uplift

Actual measurement occurs in Board 5.

Optimization Preference

Where appropriate:

Optimize For

- Conversion
- Revenue
- Margin
- Customer Relevance
- Balanced
- Business-Defined Weighted Objective

Important Rule

The AI should not independently decide that:

“Revenue is more important than customer relevance.”

The business defines the objective; AI optimizes within approved boundaries.

Actions

Save Strategy | Continue | Ask AI for Recommendation

Screen 3 — Recommendation Use Case Library

Purpose

Define the types of recommendation scenarios TICVAI supports and allow reusable recommendation use cases across venues.

Use Case Categories

Upsell

Recommend a higher-value version of what the customer is considering.

Example:

General Admission → VIP Admission

Cross-Sell

Recommend a complementary product.

Example:

Admission → Meal Package

Upgrade

Example:

Day Ticket → Annual Membership

Add-On

Example:

Admission → Guided Tour

Bundle

Example:

2 Adults + 2 Children → Family Package

Next-Best-Product

AI selects the most relevant available product.

Next-Best-Action

Example:

Renew Membership

rather than selling another ticket.

Experience Recommendation

Example:

Recommend Dolphin Encounter

F&B Recommendation

Example:

Recommend Family Meal

Retail Recommendation

Example:

Recommend Souvenir Package

Use Case Card

Each use case shows:

- Name
- Type
- Source Product / Context
- Recommended Product Category
- Objective
- Applicable Venues
- Channels
- Status

Template Library

Provide reusable templates:

Admission Upsell

Membership Upgrade

F&B Cross-Sell

Experience Add-On

Family Bundle

Post-Visit Return Offer

Clone & Adapt

A venue should be able to:

Use Template

→ customize it

rather than recreate every recommendation strategy manually.

Actions

Create Use Case | Use Template | Clone | View Dependencies

Screen 4 — Recommendation Type & Candidate Configuration

Purpose

Define **what TICVAI is allowed to recommend** for each recommendation strategy.

This creates the candidate universe before Board 3 ranks anything.

Recommendation Sources

Allow recommendation candidates from relevant TICVAI modules:

- Ticket Products
- Ticket Upgrades
- Memberships
- Subscriptions
- Passes
- Experiences
- Add-Ons
- F&B
- Retail
- Bundles
- Promotions
- Loyalty Benefits
- Other supported products/services

Candidate Configuration

Example strategy:

Family Visit Cross-Sell

Candidate Groups:

- ✓ Meal Packages
- ✓ Family Experiences
- ✓ Souvenir Bundles
- ✓ Membership

Exclude:

- ✗ Adult-only Experiences

Candidate Selection Modes

Explicit

Administrator selects specific products.

Category-Based

Example:

All Guided Tours

Rule-Based

Example:

Products tagged Family

AI Candidate Discovery

AI may identify potentially relevant products, but they must still satisfy business eligibility.

Candidate Priority

Allow:

- Preferred
- Standard
- Low Priority
- Fallback Only

Candidate Limits

Example:

Maximum candidates passed to AI ranking: 50

Maximum recommendations returned: 3

Product Availability Principle

A product should not become recommendable merely because it exists.

It must pass:

Active

-

Saleable

-

Channel Eligible

-

Date / Time Valid

-

Capacity / Inventory Available

-

Customer Eligible**Actions**

Add Candidate | Add Category | Exclude Product | Preview Candidate Pool

Screen 5 — Product Relationship & Recommendation Map

Purpose

Define and visualize business relationships between TICVAI products that can influence recommendations.

This should integrate with existing product relationships rather than create conflicting relationships.

Relationship Types

Support:

- Upsell To
- Cross-Sell With
- Upgrade To
- Downgrade To
- Complementary To
- Alternative To
- Requires
- Conflicts With
- Bundle With
- Substitute For
- Recommended After
- Recommended Before

Visual Relationship Map

Example:

General Admission

→ Upsell To → **VIP Admission**

→ Cross-Sell → **Meal Package**

→ Add-On → **Guided Tour**

→ Upgrade To → **Annual Membership**

Relationship Strength

Where useful:

- Strong
- Medium
- Weak

or numerical weight.

Example

Adult Admission

Relationship:

Upgrade To

Annual Pass

Priority:

High

Condition:

If upgrade eligibility exists.

Existing Product Relationship Integration

If Ticketing already defines:

Guided Tour REQUIRES Admission

Recommendation AI must understand this dependency automatically.

Conflict Example

If:

Product A conflicts with Product B

then Product B should not normally be recommended to a customer already purchasing Product A.

Relationship Source

Show:

- Business Configured
- Existing TICVAI Product Relationship
- AI Suggested
- Imported

Actions

Create Relationship | Edit | Remove | Ask AI to Suggest | View Product

Screen 6 — Recommendation Eligibility & Exclusion Rules

Purpose

Define the hard business rules controlling when a recommendation is permitted.

This is one of the most important screens on Board 1.

The AI ranking engine must **never override hard eligibility/exclusion rules**.

Rule Builder

Structure:

IF

conditions

THEN

Candidate Eligible / Excluded

Eligibility Dimensions

Allow supported conditions based on:

- Product Owned
- Product in Basket
- Ticket Type
- Membership
- Loyalty Tier
- Customer Segment
- Age / Eligibility
- Visit Date
- Venue
- Channel
- Sales Channel
- Capacity
- Inventory
- Product Availability
- Transaction Value
- Party Composition
- Reservation
- Time
- Other supported business attributes

Example

Fast Pass Recommendation

Eligible when:

Customer has valid Admission

AND

Fast Pass inventory > 0

AND

Fast Pass not already owned

AND

Visit date = applicable date

Exclusion

Exclude when:

Customer already owns Fast Pass

OR

Fast Pass sold out

OR

Product unavailable for selected channel

Hard vs Soft Rule

Hard

Cannot be overridden.

Example:

Age restriction.

Soft

May influence ranking.

Example:

Customer purchased similar experience previously.

Soft signals are mainly consumed by Board 3.

Rule Testing

Provide:

Test Eligibility

Input sample context.

Result:

Eligible ✓

with explanation.

Actions

Create Rule | Test Rule | Add Exclusion | Save

Screen 7 — Business Priority, Constraints & Guardrails

Purpose

Allow business teams to control AI behavior without manually selecting every recommendation.

This screen defines **what AI may optimize within**.

Business Priority

Example:

Priority Order

1. Membership Upgrade
2. Premium Experience
3. F&B
4. Retail

This does not necessarily force the final recommendation but can influence decisioning according to configured strategy.

Constraints

Configure:

- Maximum Recommendations
- Maximum Upsells
- Maximum Cross-Sells
- Minimum / Maximum Offer Value
- Margin Constraint where available
- Capacity Constraint
- Inventory Constraint
- Channel Constraint
- Time Constraint

Customer Experience Guardrails

Examples:

Do not recommend more than 3 items at checkout.

Do not recommend a lower-tier product after customer selected VIP.

Do not recommend an unavailable experience.

Do not recommend membership when customer already holds equivalent membership.

Commercial Guardrails

Examples:

Do not recommend products below a configured margin threshold where margin data is available.
Prioritize owned inventory over external inventory where policy requires.

Operational Guardrails

Example:

Stop recommending an experience when remaining capacity falls below configured operational reserve.

Strategy Priority

If multiple strategies compete for the same placement:

Membership Conversion — Priority 90

F&B Cross-Sell — Priority 70

Retail Cross-Sell — Priority 50

Actions

Add Guardrail | Change Priority | Test Conflict | Save

Screen 8 — Recommendation Conflict & Suppression Management

Purpose

Resolve situations where multiple recommendation strategies, products, or business rules compete or contradict one another.

Conflict Command Center

Header KPIs:

- Active Conflicts
- Strategy Conflicts
- Product Conflicts
- Placement Conflicts
- Rule Conflicts
- Suppressed Recommendations

Example Conflict

Strategy A

Recommend Annual Membership

Strategy B

Recommend VIP Day Upgrade

Both target:

Checkout — Primary Recommendation Slot

Resolution

Options:

- Higher Strategy Priority
- Highest Expected Value
- Highest Relevance
- Business Rule
- AI Decision within guardrails
- Suppress One
- Different Placement

Product Conflict

Example:

Customer has:

VIP Admission

System attempts to recommend:

General Admission

Result:

Suppressed

Reason:

Downgrade not permitted in this use case.

Suppression Rules

Support:

- Already Purchased
- Already Recommended
- Customer Rejected
- Sold Out
- Expired
- Incompatible Product
- Business Suppression
- Frequency Limit
- Regulatory / Eligibility
- Manual Suppression

Suppression Scope

- Customer
- Segment
- Product

- Strategy
- Channel
- Venue
- Time Period

Actions

Resolve Conflict | Create Suppression | Review Rule | Simulate

Screen 9 — Recommendation Fallback & Default Strategy

Purpose

Define what TICVAI should do when AI cannot generate an appropriate recommendation.

This prevents empty recommendation placements and unsafe AI improvisation.

Fallback Hierarchy

Example:

Level 1

AI Personalized Recommendation

↓

If unavailable:

Level 2

Rule-Based Recommendation

↓

If unavailable:

Level 3

Business-Priority Product

↓

If unavailable:

Level 4

Popular / Trending Eligible Product

↓

If unavailable:

Level 5

No Recommendation

Confidence Threshold

Example:

AI recommendation confidence:

0.78

Minimum required:

0.70

Result:

Use AI Recommendation

If:

0.54

Result:

Use Fallback Strategy

Default Strategies

Examples:

Most Popular Eligible

Business Featured

Highest Priority Eligible

Recently Popular

No Recommendation

Cold Start

For:

- New Customer
- Anonymous Customer
- New Product
- New Venue
- Insufficient History

Configure fallback.

Example:

Anonymous Visitor

→ Context-Based

→ Popular Eligible Products

Failure Handling

If recommendation service unavailable:

Use Rule-Based Fallback.

If candidate data unavailable:

Return no recommendation rather than fabricate one.

Actions

[Configure Fallback](#) | [Change Threshold](#) | [Test Scenario](#) | [Save](#)

Screen 10 — Recommendation Strategy Preview & Simulation

Purpose

Validate the complete recommendation strategy before activating it or passing it to later recommendation boards.

This is the final output of Board 1.

Strategy Header

Fast Pass & Experience Upsell Strategy

Status:

Ready for Activation

Configuration Completion:

100%

Strategy Summary

Objective

Increase Average Order Value

Primary Use Case

Upsell / Cross-Sell

Candidate Groups

Fast Pass
Premium Experiences
Meal Packages

Channels

B2C
POS
Mobile App

Maximum Recommendations

3

Decision Mode

AI Ranking + Business Rules

Simulation Input

Create a test context.

Example:

Customer

Family Visitor

Party

2 Adults + 2 Children

Current Basket

4 × General Admission

Visit

Saturday

Venue

Theme Park

Channel

B2C

Candidate Funnel

This visualization is important.

24 Potential Products

↓

18 Active & Saleable

↓

12 Channel Eligible

↓

8 Customer Eligible

↓

5 Pass Business Rules

↓

3 Passed to AI Ranking**Preview Output****#1 Fast Pass Upgrade**

Eligible ✓

Business Priority: High

AI Ranking:

Board 3 will calculate final intelligence.

#2 Family Meal Package

Eligible ✓

Priority: Medium

#3 Annual Membership Upgrade

Eligible ✓

Priority: Medium

Explainability Preview

Show why each candidate survived:

Fast Pass

- ✓ Admission owned
- ✓ Saturday eligible
- ✓ Capacity available
- ✓ Not already owned
- ✓ Correct channel

Configuration Issues

Example:

0 Blocking

2 Recommendations

1 Warning

Actions

Run Another Simulation | Modify Strategy | Save Draft

Primary CTA

Activate Strategy Foundation / Continue to Audience Intelligence →

This moves the configuration into Board 2.

Critical Recommendation Pipeline

Soft Labs should understand that recommendation generation is a **pipeline**, not a single AI prompt.

The future runtime architecture should follow:

Customer / Anonymous Visitor

-

Current Context

-

Journey Stage

↓

Strategy Selection

Which recommendation strategies are applicable?

↓

Candidate Generation

What could theoretically be recommended?

↓

Hard Eligibility

What is actually allowed?

↓

Exclusions / Suppression

What must not be recommended?

↓

Business Constraints

What remains commercially and operationally acceptable?

↓

AI Decisioning & Ranking

Board 3

↓

Channel Delivery

Board 4

↓

Measurement

Board 5

This architecture is much safer and more controllable than asking an LLM:

“What should I sell this customer?”

Hard Rules vs AI Intelligence

This separation should be explicit.

Hard Rule

Customer must be at least 18.

AI cannot override it.

Availability Rule

Product must have capacity.

AI cannot override it.

Business Rule

Don't recommend Fast Pass if already owned.

AI cannot override it.

AI Intelligence

Among the **eligible remaining candidates**, AI determines:

Which recommendation is most relevant to this customer/context?

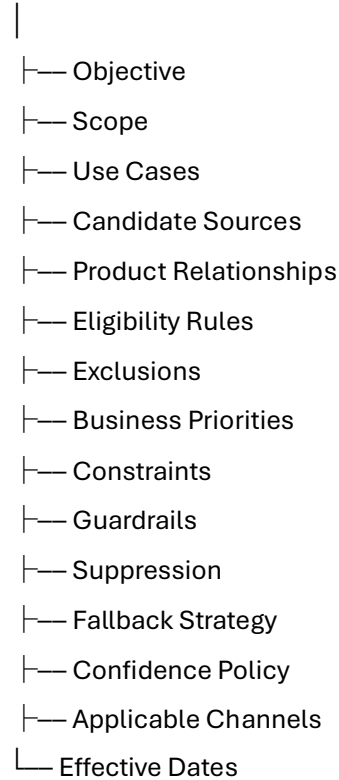
That is the correct relationship between **business governance and AI intelligence**.

Recommendation Object Model

I recommend TICVAI maintain a structured Recommendation Strategy object.

Conceptually:

Recommendation Strategy



This object becomes the business foundation consumed by Boards 2–5.

AI Suggestions During Configuration

AI can help the administrator configure recommendation strategies.

For example:

“You selected Membership Conversion as the objective. Would you like me to identify ticket products that could logically upgrade to Annual Membership?”

or:

“Fast Pass has limited capacity. I recommend adding an availability guardrail so it stops being recommended below your operational reserve.”

But AI suggestions must be visibly classified as:

AI Recommended

and require appropriate confirmation.

Relationship with Existing TICVAI Modules

Board 1 should **reference**, not duplicate, existing configuration.

Ticketing

Provides products and product relationships.

Pricing

Provides current price.

Dynamic Pricing

Provides applicable dynamic price.

Promotions

Provides promotions/discounts.

Membership

Provides membership products and eligibility.

Loyalty

Provides loyalty status/benefits.

Capacity

Provides availability.

F&B / Retail

Provide sellable products and inventory.

CRM

Provides customer information where permitted.

The Recommendation module consumes these capabilities to determine what may be recommended.

Important Boundary with Promotions

A recommendation is not necessarily a promotion.

For example:

“Upgrade to VIP Admission.”

Recommendation Engine determines:

VIP is relevant.

Pricing determines:

VIP costs AED 250.

Promotion determines:

Customer receives 10% discount, if applicable.

These responsibilities should remain separate.

Important Boundary with Dynamic Pricing

Similarly:

Recommendation Engine:

Recommend Fast Pass.

Dynamic Pricing:

Current Fast Pass price = AED 95.

The Recommendation Engine should **consume AED 95**, not calculate its own dynamic price.

Board 1 — Final Screen Map

Recommendation Strategy & Configuration

1. Recommendation Strategy Command Center
2. Recommendation Objective & Strategy Builder
3. Recommendation Use Case Library
4. Recommendation Type & Candidate Configuration
5. Product Relationship & Recommendation Map
6. Recommendation Eligibility & Exclusion Rules
7. Business Priority, Constraints & Guardrails
8. Recommendation Conflict & Suppression Management
9. Recommendation Fallback & Default Strategy
10. Recommendation Strategy Preview & Simulation

This gives us a clean foundation. **Board 1 controls what TICVAI is trying to achieve and what it is allowed to recommend. Board 2 should now focus entirely on understanding the customer, audience, behavior, and real-time context before Board 3 performs AI decisioning and ranking.**

TICVAI — Scope of Work

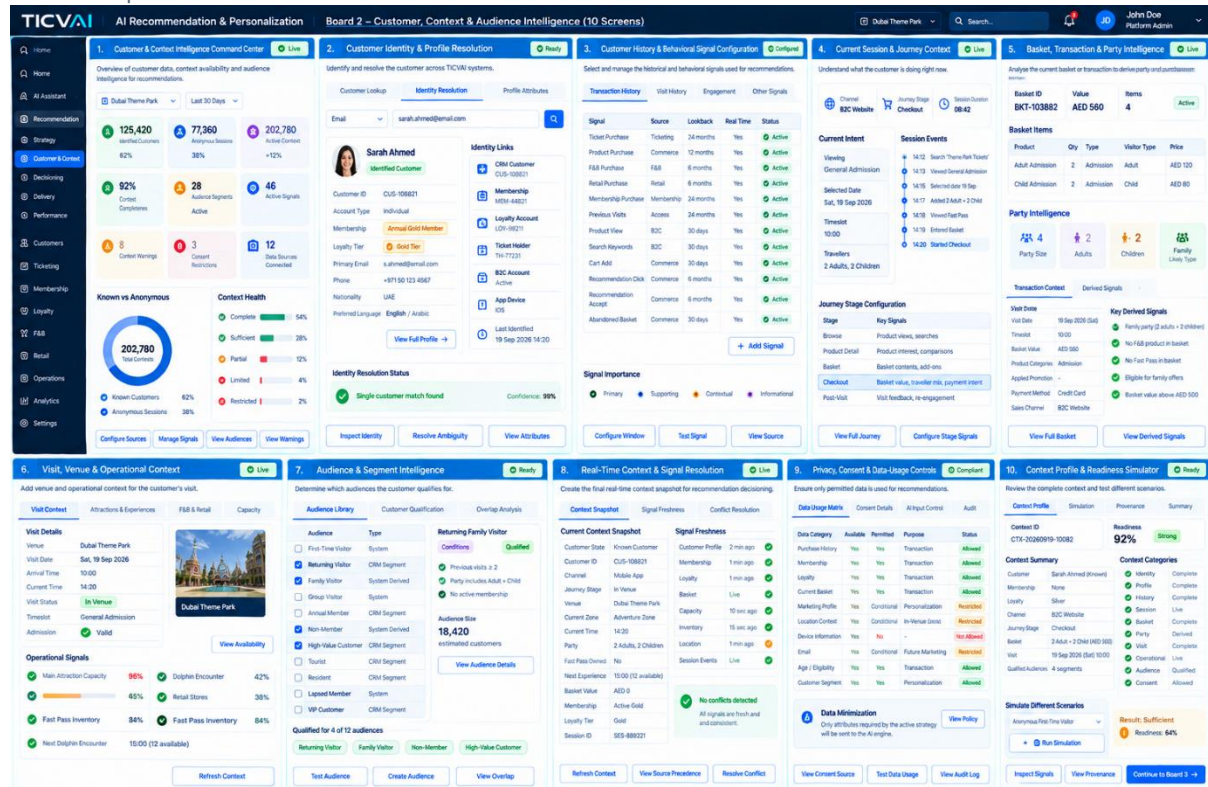
Recommendation & Personalization — Board 2: Customer, Context & Audience Intelligence

Module: AI Recommendation & Personalization

Board: 2 of 5

Screens: 10

Board Type: Customer Intelligence / Context Resolution / Audience / Behavioral Signals / Journey Context / Real-Time Decision Inputs



Board Purpose

Board 1 answered:

What are we trying to recommend, and what are the business rules?

Board 2 answers:

Who is the customer, what is happening right now, and what information can TICVAI legitimately use to personalize the recommendation?

This board creates the **Recommendation Context** that will later be consumed by Board 3 — Recommendation Decisioning, Ranking & AI Intelligence.

TICVAI must support both:

Known Customer → customer/account/member information is available.

and:

Anonymous Visitor → TICVAI may still personalize using permitted session and transaction context without requiring identification.

The architecture should be:

Customer Identity + Profile + Membership + Loyalty + History + Current Session + Basket + Party + Visit + Venue + Channel + Journey Stage + Operational Context

↓

Recommendation Context

↓

Audience / Segment Qualification

↓

Board 3 — Candidate Scoring & Ranking

A critical principle: **Board 2 does not choose the recommendation.** It assembles, qualifies, and explains the context used by the recommendation engine.

Screen 1 — Customer & Context Intelligence Command Center

Purpose

Provide one central operational and configuration view of the information available to the recommendation engine.

Header KPIs

- Identified Customers
- Anonymous Sessions
- Active Recommendation Contexts
- Context Completeness
- Audience Segments
- Active Signals
- Real-Time Signals
- Context Warnings
- Consent / Usage Restrictions
- Data Sources Connected

Context Coverage

Show coverage by intelligence category:

Intelligence Area	Source	Availability	Freshness	Status
Customer Profile	CRM	94%	Real Time	Active
Purchase History	Ticketing	100%	Real Time	Active
Membership	Membership	100%	Real Time	Active
Loyalty	Loyalty	100%	Real Time	Active
Current Basket	Commerce	100%	Live	Active
Visit Context	Ticketing	98%	Live	Active
Capacity	Capacity	100%	Live	Active

Intelligence Area	Source	Availability	Freshness	Status
F&B History	F&B	82%	Near Real Time	Warning

Known vs Anonymous

Visual breakdown:

Known Customers — 62%

Anonymous — 38%

TICVAI must make it clear that recommendation intelligence can operate in both scenarios, with different available signals.

Context Health

Statuses:

- Complete
- Sufficient
- Partial
- Limited
- Restricted
- Unavailable

Actions

[Configure Sources](#) | [Manage Signals](#) | [View Audiences](#) | [Inspect Context](#) | [View Warnings](#)

Screen 2 — Customer Identity & Profile Resolution

Purpose

Determine **who the recommendation engine is dealing with** and resolve available customer identities without creating duplicate customer records.

This screen consumes CRM/customer identity capabilities rather than becoming a second CRM.

Identity States

Support:

- Anonymous Visitor
- Recognized Visitor
- Logged-In Customer
- Ticket Holder
- Member
- Loyalty Member
- Corporate Customer
- Group Booker
- Guest / Companion

Resolved Customer Context

Example:

Customer

Sarah Ahmed

Customer ID

CUS-108821

Status

Identified

Membership

Annual Gold

Loyalty

Gold Tier

Active Tickets

4

Upcoming Visits

1

Identity Signals

Possible permitted signals:

- Customer ID
- Account ID
- Membership ID
- Loyalty ID
- Ticket Holder ID
- Booking Customer
- Session ID
- Basket ID
- Device/session reference where permitted
- Authenticated app session

Identity Resolution

Example:

B2C Session

↓

Logged-In Account

↓

CRM Customer

↓

Membership



Loyalty Account



Unified Recommendation Identity

Duplicate / Ambiguous Identity

If TICVAI finds:

2 possible customer records

it should not silently merge them.

Status:

Identity Resolution Required

Profile Attributes Available to Recommendation

Display allowed attributes such as:

- Customer Type
- Preferred Language
- Country / Market where permitted
- Membership
- Loyalty Tier
- Customer Segment
- Account Type
- Relationship to booking

Sensitive or restricted information must not automatically become recommendation features.

Actions

[View Customer](#) | [Inspect Identity](#) | [Resolve Ambiguity](#) | [View Available Attributes](#)

Screen 3 — Customer History & Behavioral Signal Configuration

Purpose

Configure which historical and behavioral signals may be supplied to Recommendation Intelligence.

Signal Categories

Transaction History

- Tickets Purchased
- Products Purchased
- Membership Purchases
- Upgrades
- Add-Ons

- F&B Purchases
- Retail Purchases
- Refunds
- Exchanges
- Reschedules

Visit History

- Previous Visits
- Visit Frequency
- Last Visit
- Attractions Used
- Experiences Attended
- Entitlements Redeemed

Engagement

Where available:

- Products Viewed
- Searches
- Cart Adds
- Cart Removes
- Abandoned Basket
- Recommendation Click
- Recommendation Accepted
- Recommendation Rejected

Signal Configuration Table

Signal	Source	Window	Weight Use	Real Time	Status
Ticket Purchase	Ticketing	24 Months	Yes	Yes	Active
Visit History	Access	12 Months	Yes	Yes	Active
F&B Purchase	F&B	6 Months	Yes	Yes	Active
Product View	B2C	30 Days	Yes	Yes	Active

Lookback Windows

Configurable per signal:

- Current Session
- 7 Days
- 30 Days
- 90 Days
- 6 Months

- 12 Months
- Custom

Signal Importance

Board 2 may classify signals:

- Primary
- Supporting
- Contextual
- Informational

But **final scoring weights belong to Board 3.**

Missing Data

Example:

No previous purchase history is available.

This should become:

Signal Unavailable

not:

Customer is not interested.

Actions

Add Signal | Configure Window | Disable Signal | View Source | Test Signal

Screen 4 — Current Session & Journey Context

Purpose

Understand **what the customer is doing right now.**

For many recommendations, current intent may be more important than historical behavior.

Current Session Context

Example:

Channel

B2C

Journey Stage

Checkout

Current Venue

Dubai Theme Park

Session Duration

08:42

Customer State

Authenticated

Basket Value

AED 560

Journey Stages

Support configurable stages such as:

- Homepage
- Browse
- Search
- Product Detail
- Product Selection
- Date Selection
- Timeslot Selection
- Seat Selection
- Basket
- Checkout
- Payment
- Confirmation
- Pre-Visit
- Arrival
- In-Venue
- Exit
- Post-Visit
- Renewal

Current Intent Signals

Examples:

Viewing

General Admission

Selected Date

Saturday 19 September

Current Basket

2 Adult + 2 Child

Removed

Fast Pass

Viewed

Annual Membership

Session Events

Timeline example:

09:02 Search "Theme Park Tickets"

09:04 Viewed General Admission

09:06 Selected Saturday

09:08 Added 2 Adult + 2 Child

09:10 Viewed Fast Pass

09:11 Entered Basket

Journey Context Configuration

Administrators can define which signals are meaningful at different stages.

For example:

Checkout

Basket context = High relevance

Browsing history = Supporting

Long-term history = Supporting

Actions

[View Session](#) | [Configure Journey Signals](#) | [Inspect Events](#) | [Test Context](#)

Screen 5 — Basket, Transaction & Party Intelligence

Purpose

Transform the current basket or transaction into structured recommendation context.

This is especially important for ticketing, attractions and family/group purchases.

Basket Summary

Example:

Basket ID

BKT-103882

Value

AED 560

Items

4

Visit Date

Saturday

Basket Items

Product	Qty	Type	Visitor
Adult Admission	2	Admission	Adult
Child Admission	2	Admission	Child

Party Intelligence

Derived structure:

Party Size

4

Adults

2

Children

2

Party Type

Likely Family

The UI must distinguish:

Explicit Information

from:

Derived Context

For example:

“Likely Family” = System Derived

not a confirmed personal relationship unless the booking actually contains that information.

Transaction Context

Include:

- Basket Value
- Product Mix
- Quantity
- Admission Type
- Visit Date
- Visit Time
- Number of Guests
- Product Categories
- Existing Add-Ons
- Applied Promotion
- Membership Usage
- Voucher Usage
- Sales Channel

Missing Opportunity Detection

The system may identify contextual facts such as:

Admission present, no F&B product.

or:

Four admission tickets, no group/family package selected.

These are **signals**, not yet recommendations.

Actions

[Inspect Basket](#) | [View Party Context](#) | [View Derived Signals](#) | [Refresh Context](#)

Screen 6 — Visit, Venue & Operational Context

Purpose

Add the operational reality of the customer's intended or current visit to Recommendation Context.

This prevents AI from making recommendations that are theoretically relevant but operationally inappropriate.

Visit Context

Example:

Venue

Dubai Theme Park

Visit Date

Saturday 19 September

Arrival

10:00

Current Time

14:20

Visit Status

In Venue

Operational Signals

Consume information from:

- Operating Calendar
- Schedule
- Timeslots
- Capacity
- Inventory
- Access Control
- Resource Management
- Virtual Queue
- F&B
- Retail
- Entertainment / Attractions

Example

Customer is currently in venue.

TICVAI knows:

Dolphin Encounter

Next session: 15:00

Availability: 12

Distance / zone context where supported: Relevant

Customer eligibility: Valid

This information can later be consumed by Board 3.

Availability Context

Show:

- Available
- Limited
- Sold Out
- Closed
- Temporarily Unavailable
- Not Applicable

Operational Pressure

Example:

Main Attraction

Capacity Utilization: 96%

Secondary Attraction

Capacity Utilization: 42%

This can become a ranking input if the business permits utilization-based recommendations.

Important Boundary

Board 2 reads operational context.

It does **not** change:

- Capacity
- Price
- Queue Priority
- Operating Schedule
- Access Rules

Actions

View Venue Context | View Availability | Inspect Operational Signals | Refresh

Screen 7 — Audience & Segment Intelligence

Purpose

Determine which configured audiences or segments the current customer/context qualifies for.

This screen should integrate with CRM segmentation rather than rebuild the entire CRM segmentation engine.

Audience Library

Examples:

- First-Time Visitor
- Returning Visitor
- Annual Member
- Family Visitor
- Group Visitor
- High-Value Customer
- Tourist
- Resident
- Frequent Visitor
- Lapsed Member
- VIP Customer
- Anonymous Visitor

Audience Definition

Example:

Returning Family Visitor

Conditions:

Previous Visits ≥ 2

AND

Current Party includes Adult + Child ticket types

AND

No Active Membership

Audience Types

- CRM Segment
- Recommendation-Specific Audience
- Real-Time Context Audience
- System Derived
- Imported Audience

Membership Display

Customer may qualify for multiple audiences:

- ✓ Returning Visitor
- ✓ Family Context
- ✓ High Basket Value
- ✓ Non-Member

Priority / Relevance

Audience priority can be configured for recommendation usage.
However, final recommendation scoring remains Board 3.

Audience Size Preview

Where data is available:

Estimated Eligible Customers

18,420

Overlap Analysis

Example:

Family Visitor

n

Annual Member

= 3,842 customers

Actions

[Create Audience](#) | [Import Segment](#) | [Test Audience](#) | [View Overlap](#) | [View Members](#)

Screen 8 — Real-Time Context & Signal Resolution

Purpose

Create the final **point-in-time context snapshot** that Board 3 will use when making a recommendation.

Historical information alone is insufficient.

TICVAI must understand:

What is true right now?

Real-Time Context Snapshot

Example:

Customer

Known

Channel

Mobile App

Journey

In Venue

Venue

Theme Park

Current Zone

Adventure Zone

Time

14:20

Party

Family of 4

Admission

Valid

Fast Pass

Not Owned

Next Available Experience

15:00

Basket

None

Signal Freshness

Every real-time signal should include freshness.

Example:

Signal	Value	Updated	Status
Attraction Capacity	42%	5 sec ago	Fresh
Fast Pass Inventory	84	3 sec ago	Fresh
Current Basket	AED 0	Live	Fresh
Customer Segment	Family	2 hrs ago	Valid

Freshness Rules

Configure:

- Real Time
- < 1 Minute
- < 5 Minutes
- < 15 Minutes
- Session
- Daily
- Historical

Stale Signal Handling

If:

Capacity data last updated 45 minutes ago

TICVAI may mark:

Stale — Do Not Use for Real-Time Recommendation

depending on policy.

Signal Conflict

Example:

CRM says:

Non-Member

Membership service says:

Active Gold Member

Resolution policy:

Membership Service = Authoritative

Source Precedence

Allow source-of-truth configuration.

Example:

Membership

1. Membership Service
2. CRM
3. Imported Data

Actions

Refresh Context | Configure Freshness | Resolve Conflict | View Source

Screen 9 — Context Privacy, Consent & Data-Usage Controls

Purpose

Ensure Recommendation Intelligence uses only data that TICVAI is permitted to use.

This screen should **consume TICVAI's privacy/consent governance** rather than create a second consent-management module.

Data Usage Matrix

Data Category	Available	Permitted for Recommendation	Source	Status
Purchase History	Yes	Yes	Ticketing	Allowed
Membership	Yes	Yes	Membership	Allowed

Data Category	Available	Permitted for Recommendation	Source	Status
Current Basket	Yes	Yes	Commerce	Allowed
Marketing Profile	Yes	Conditional	CRM	Restricted
Location Context	Yes	Conditional	App	Restricted

Usage States

- Allowed
- Restricted
- Consent Required
- Purpose Restricted
- Unavailable
- Expired
- Not Applicable

Purpose Control

Example:

Data may be allowed for:

✓ Transaction Recommendation

but not:

✗ Marketing Retargeting

where applicable.

Anonymous Context

TICVAI should still support:

Session-based recommendations

without forcing customer identification.

AI Input Control

Show exactly which context categories are permitted to be passed to AI decisioning.

Example:

✓ Current Basket

✓ Product History

✓ Membership Tier

✓ Visit Context

✗ Restricted Attribute

Data Minimization

Provide policy:

Only send attributes required by the active recommendation strategy.

Do not send the entire CRM customer profile to the AI engine unnecessarily.

Audit

Record:

- Data categories used
- Recommendation session
- Purpose
- Source
- Permission status
- Policy applied

Actions

[View Policy](#) | [View Consent Source](#) | [Test Eligibility](#) | [View Audit](#)

Screen 10 — Recommendation Context Profile & Readiness Simulator

Purpose

Combine everything from Board 2 into one structured **Recommendation Context Profile** and verify that sufficient information is available before sending it to Board 3.

This is the final output of Board 2.

Context Header

Recommendation Context

CTX-20260919-10082

Customer:

Sarah Ahmed

State:

Known Customer

Channel:

B2C

Journey:

Checkout

Context Readiness:

92% — Strong

Context Categories

Category	Status	Signals
Identity	Complete	5
Profile	Complete	8

Category	Status	Signals
History	Complete	16
Session	Live	12
Basket	Complete	9
Party	Derived	4
Visit	Complete	7
Operational	Live	11
Audience	Qualified	4
Consent	Allowed	8

Context Summary

Customer

Returning Visitor

Membership

None

Loyalty

Silver

Current Basket

2 Adult + 2 Child Admission

Party Context

Likely Family

Visit

Saturday

Channel

B2C

Journey Stage

Checkout

Qualified Audiences

- ✓ Returning Visitor
- ✓ Family Context
- ✓ Non-Member
- ✓ High Basket Value

Real-Time Signals

- ✓ Fast Pass Available
- ✓ Family Meal Available
- ✓ Guided Tour Available
- ✓ Annual Membership Active
- ✓ Venue Open
- ✓ Capacity Data Fresh

Restricted Signals

Example:

Precise Location

Not permitted for this recommendation context.

Therefore:

Excluded from AI Input

Context Provenance

Every signal must identify its origin:

- Customer Provided
- Transaction Derived
- System Derived
- CRM
- Membership
- Loyalty
- Ticketing
- Access
- Commerce
- Operational System
- Imported

Simulation

Allow the administrator to test different scenarios.

Scenario A

Known Returning Family

Result:

Context Readiness — 92%

Scenario B

Anonymous First-Time Visitor

Available:

- Channel

- Basket
- Visit
- Product
- Party Context
- Operational Context

Unavailable:

- Customer History
- Membership History
- Loyalty

Result:

Context Readiness — 64% / Sufficient

Board 3 can still generate recommendations using available context and fallback logic.

Scenario C

Known Member In-Venue

Available:

- Membership
- History
- Access
- Venue
- Current Time
- Operational Availability

Result:

Context Readiness — 96%

Actions

Run Simulation | Change Context | Inspect Signals | View Provenance

Primary CTA

Continue to AI Decisioning & Ranking →

Critical Board 2 Architecture

Soft Labs should implement Board 2 around a normalized **Recommendation Context Object**.

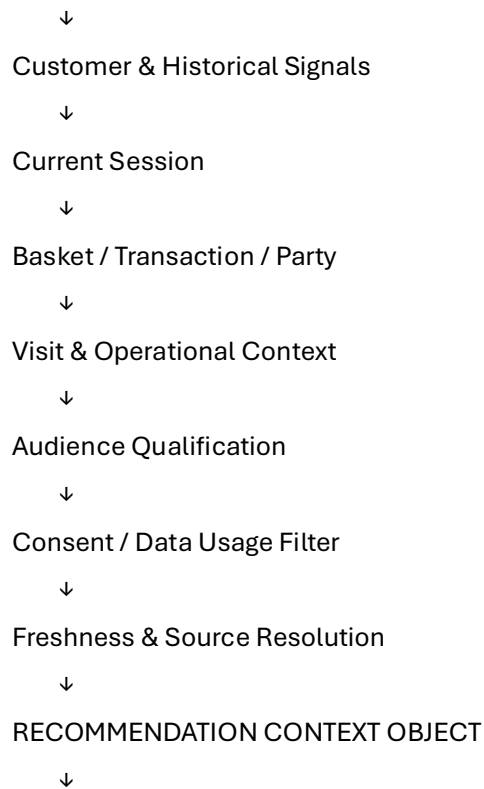
It should not allow Board 3 to independently query random customer information every time it wants to make a recommendation.

The flow should be:

Customer / Session

↓

Identity Resolution



Board 3 — Decisioning & Ranking

This provides a controlled interface between TICVAI's operational systems and the AI layer.

Recommendation Context Object

I recommend the technical object conceptually contain:

RecommendationContext



- └─ Basket Context
- └─ Party Context
- └─ Visit Context
- └─ Operational Context
- └─ Qualified Audiences
- └─ Real-Time Signals
- └─ Data Permissions
- └─ Signal Provenance
- └─ Signal Freshness
- └─ Context Readiness

Board 3 consumes this object rather than assembling customer information itself.

Signal Provenance — Mandatory

This is particularly important for AI explainability.

Every signal should carry:

Value + Source + Timestamp + Confidence/Status + Permission

For example:

Signal:

Party Type

Value:

Family Context

Source:

System Derived

Derived From:

2 Adult + 2 Child tickets

Timestamp:

12 Sep 2026 14:20

Status:

Valid

This prevents TICVAI from presenting an inference as if the customer explicitly told us something.

Anonymous Customers

Do not design the recommendation system assuming a CRM customer must exist.

For an anonymous B2C visitor TICVAI can still have:

Anonymous Session

↓

Viewed Product

↓

Selected Date

↓

Basket

↓

Party Composition

↓

Venue

↓

Channel

↓

Journey Stage

↓

Operational Availability

↓

Context-Based Recommendation

If the customer later authenticates, the context can be enriched rather than discarded.

Context Hierarchy

For real-time recommendations I recommend Board 3 generally receive context in this conceptual hierarchy:

1. Hard Eligibility & Permission

Can this information/product legally and operationally be used?

↓

2. Current Intent

What is the customer doing now?

↓

3. Current Transaction

What are they purchasing?

↓

4. Visit Context

Where/when is the experience happening?

↓

5. Customer Relationship

Member, loyalty, returning customer, etc.

↓

6. Historical Behavior

What happened previously?

This prevents historical behavior from overwhelming obvious current intent.

The actual weighting, however, belongs to **Board 3**.

Important Boundary — CRM

Board 2 must not duplicate CRM.

CRM owns:

- Customer Master
- Customer Profiles
- CRM Segments
- Communication Preferences
- Customer History where applicable

Recommendation Intelligence **consumes** those objects.

It can create recommendation-specific real-time audiences, but it should not become TICVAI's second customer database.

Important Boundary — Reporting & Analytics

Board 2 may display **context availability and signal health**, but it should not become a customer analytics dashboard.

Detailed:

- Customer analytics
- Revenue analytics
- Recommendation conversion
- Attribution
- Recommendation performance

belong to their respective reporting areas and **Board 5**.

Important Boundary — Board 3

Board 2 must **not** perform:

- Candidate AI scoring
- Recommendation ranking
- Next-best-product calculation
- Next-best-action calculation
- Model selection

- Recommendation confidence scoring
- AI explanation generation
- Recommendation optimization

Those belong to:

Board 3 — Recommendation Decisioning, Ranking & AI Intelligence

Board 2 only produces the **trusted context** Board 3 needs.

Board 2 — Final Screen Map

Customer, Context & Audience Intelligence

- 1. Customer & Context Intelligence Command Center**
- 2. Customer Identity & Profile Resolution**
- 3. Customer History & Behavioral Signal Configuration**
- 4. Current Session & Journey Context**
- 5. Basket, Transaction & Party Intelligence**
- 6. Visit, Venue & Operational Context**
- 7. Audience & Segment Intelligence**
- 8. Real-Time Context & Signal Resolution**
- 9. Context Privacy, Consent & Data-Usage Controls**
- 10. Recommendation Context Profile & Readiness Simulator**

This gives us a strong separation: **Board 1 defines the recommendation strategy and guardrails; Board 2 builds the trusted customer/context picture; Board 3 becomes the actual AI brain that scores and ranks the eligible recommendations.**

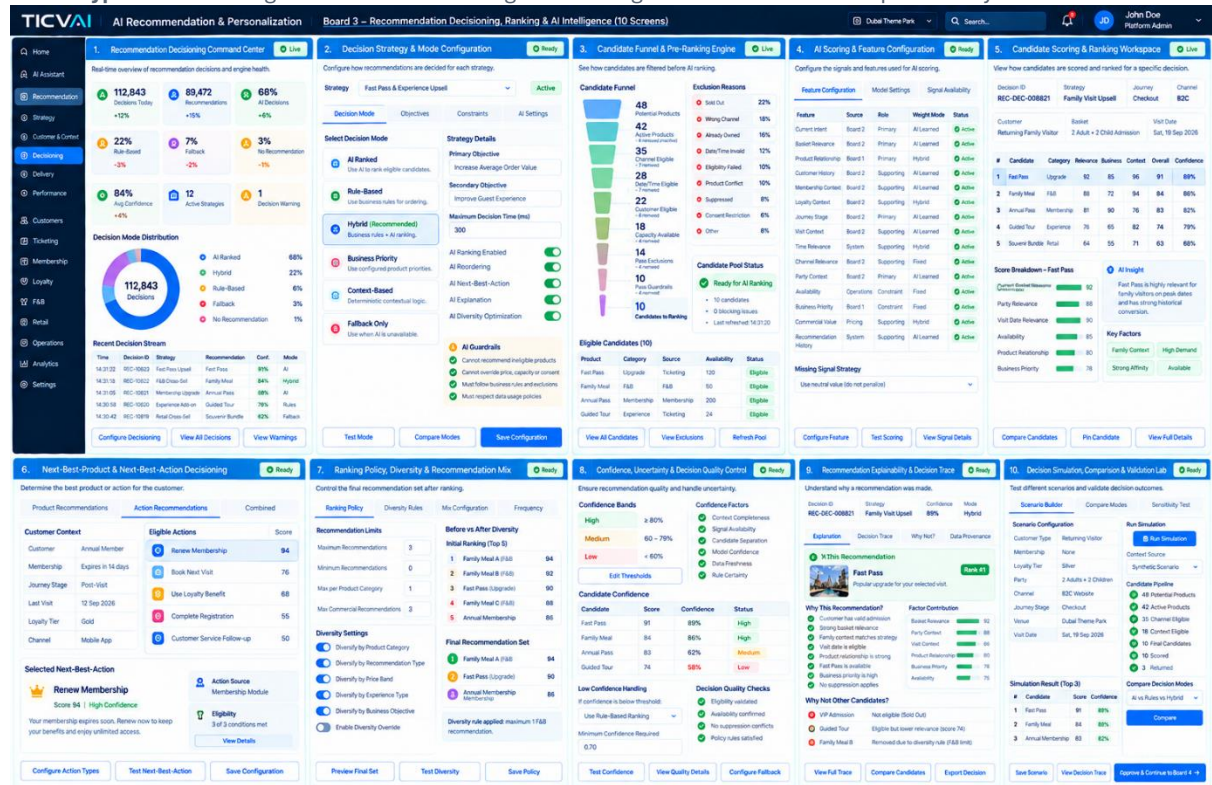
Recommendation & Personalization — Board 3: Recommendation Decisioning, Ranking & AI Intelligence

Module: AI Recommendation & Personalization

Board: 3 of 5

Screens: 10

Board Type: Decision Engine / Candidate Scoring / AI Ranking / Next-Best-Action / Explainability / Simulation



Board Purpose

Board 1 established **what TICVAI is allowed and trying to recommend**.

Board 2 established **who the customer is and what is happening right now**.

Board 3 becomes the actual **recommendation decision brain**:

From all eligible candidates, which recommendation—or action—is best for this customer, in this context, at this moment?

The runtime should follow:

Recommendation Strategy — Board 1

+

Recommendation Context — Board 2

+

Eligible Candidate Products / Actions

↓

Decisioning Engine

↓

Rules & Constraints

↓

AI Scoring

↓

Ranking

↓

Confidence & Quality Checks

↓

Next-Best-Product / Next-Best-Action

↓

Explanation

↓

Board 4 — Channel & Journey Delivery

The most important architectural rule is:

AI ranks only what TICVAI has already determined is eligible.

AI must never bypass eligibility, availability, access, capacity, commercial, consent, or other hard rules.

Screen 1 — Recommendation Decisioning Command Center

Purpose

Provide one centralized view of the Recommendation Decision Engine, its active strategies, decision volume, AI/rule usage, confidence health, and decision quality.

Header KPIs

- Decisions Today
- Recommendations Generated
- AI Decisions
- Rule-Based Decisions
- Fallback Decisions
- No-Recommendation Decisions
- Average Confidence
- Active Decision Strategies
- Decision Warnings
- Decision Failures

Decision Mode Distribution

Visualize:

AI Ranked

Rule Ranked

Hybrid

Fallback

No Recommendation

Example:

Mode	Decisions	%
AI Ranked	84,220	68%
Hybrid	27,840	22%
Rule-Based	7,180	6%
Fallback	3,410	3%
None	1,240	1%

Active Decision Strategies

Strategy	Objective	Candidates	Decision Mode	Avg Confidence	Status
Fast Pass Upsell	AOV	8	Hybrid	87%	Healthy
Family Cross-Sell	Attach Rate	12	AI	82%	Healthy
Membership Upgrade	Conversion	5	AI	91%	Healthy
F&B Cross-Sell	F&B Spend	24	Hybrid	73%	Warning

Decision Health

Statuses:

- Healthy
- Low Confidence
- Excessive Fallback
- Candidate Shortage
- Context Limited
- Rule Conflict
- Model Unavailable
- Decision Blocked

Recent Decision Stream

Show:

- Decision ID
- Context ID

- Strategy
- Candidate Count
- Selected Recommendation
- Confidence
- Decision Mode
- Latency
- Result

Actions

[Configure Decisioning](#) | [Inspect Decision](#) | [Test Strategy](#) | [View Models](#) | [View Warnings](#)

Screen 2 — Decision Strategy & Mode Configuration

Purpose

Configure **how TICVAI should make the decision** for each recommendation strategy.

Not every recommendation requires the same level of AI.

Decision Modes

1. Rule-Based

Business rules determine ordering.

Example:

Membership Upgrade always before F&B.

2. AI Ranked

AI/model ranks eligible candidates.

3. Hybrid

Business constraints + AI ranking.

This should be the recommended default for most sophisticated TICVAI use cases.

4. Business Priority

Use configured priorities from Board 1.

5. Context-Based

Use deterministic contextual logic.

Example:

In-venue + lunch period → eligible F&B candidates.

6. Fallback Only

Used when the main decision engine is unavailable.

Strategy Configuration

Example:

Strategy

Fast Pass & Experience Upsell

Decision Mode

Hybrid

Primary Objective

Increase AOV

Secondary Objective

Customer Relevance

AI Autonomy

Configure:

- AI Ranking Enabled
- AI Candidate Reordering
- AI Next-Best-Action
- AI Explanation
- AI Confidence
- AI Diversity Optimization

AI should **not** be allowed to:

- Add ineligible products
- Override hard exclusions
- Change prices
- Create promotions
- Ignore consent
- Ignore capacity

Decision Timeout

Example:

Maximum Decision Time: 300 ms

If exceeded:

→ Use configured fallback.

Actions

Save Decision Strategy | Test Mode | Compare Modes | Continue

Screen 3 — Candidate Funnel & Pre-Ranking Engine

Purpose

Show exactly how TICVAI moves from the original candidate universe into the final candidate set that AI is allowed to score.

This screen is critical for explainability.

Candidate Funnel

Example:

48 Potential Products

↓

42 Active Products

↓

35 Channel Eligible

↓

28 Date/Time Eligible

↓

22 Customer Eligible

↓

18 Capacity/Inventory Available

↓

14 Pass Exclusions

↓

10 Pass Business Guardrails

↓

10 Candidates Sent to Ranking

Candidate Table

Candidate	Source	Eligibility	Availability	Guardrails	Final
Fast Pass	Ticketing	Pass	Available	Pass	Included
Family Meal	F&B	Pass	Available	Pass	Included
Annual Pass	Membership	Pass	Available	Pass	Included
VIP Upgrade	Ticketing	Pass	Sold Out	—	Excluded
Adult Experience	Ticketing	Age Conflict	—	—	Excluded

Exclusion Reason

Every removed candidate must have a reason.

Examples:

- Already Owned
- Sold Out
- Wrong Channel
- Date Invalid

- Capacity Unavailable
- Eligibility Failed
- Conflict
- Suppressed
- Consent Restriction
- Business Guardrail

AI Boundary

The AI receives only:

10 Eligible Candidates

not all 48 products.

Candidate Refresh

Because operational context can change:

Refresh Candidate Pool

should rerun eligibility before decisioning.

Actions

Inspect Candidate | View Exclusion | Refresh Pool | Continue to Scoring

Screen 4 — AI Scoring & Feature Configuration

Purpose

Configure the signals/features the decision engine can use to score eligible candidates.

Board 2 provides the context.

Board 3 determines **how that context influences recommendation relevance**.

Scoring Dimensions

Possible dimensions:

- Current Intent
- Basket Relevance
- Product Affinity
- Customer History
- Membership Context
- Loyalty Context
- Journey Stage
- Visit Context
- Time Relevance
- Channel Relevance
- Party Context
- Product Relationship

- Availability
- Capacity Context
- Business Priority
- Commercial Value
- Customer Relevance
- Recommendation History

Feature Configuration

Feature	Source	Role	Weight Mode	Status
Basket Relevance	Board 2	Primary	AI Learned	Active
Product Relationship	Board 1	Primary	Hybrid	Active
Current Intent	Board 2	Primary	AI Learned	Active
Business Priority	Board 1	Constraint	Fixed	Active
Visit Context	Board 2	Supporting	AI Learned	Active

Weighting Modes

Support:

- AI Learned
- Business Fixed
- Hybrid
- Informational
- Disabled

Important Separation

A business administrator may say:

Membership Conversion is strategically important.

But should not necessarily need to manually configure:

Basket relevance = 0.27

Purchase history = 0.18

Visit context = 0.14

AI/model configuration should handle this where appropriate.

Feature Availability

If a feature is unavailable:

Purchase History — Missing

the system must use the configured missing-signal strategy.

It must not assume a zero preference.

Actions

[Configure Features](#) | [Disable Feature](#) | [View Source](#) | [Test Scoring](#)

Screen 5 — Candidate Scoring & Ranking Workspace

Purpose

Show how the eligible candidates are scored and ordered for a specific recommendation decision.

Decision Example

Decision ID

REC-DEC-008821

Customer Context

Returning Family Visitor

Journey

Checkout

Basket

2 Adult + 2 Child Admission

Strategy

Family Visit Upsell

Ranked Candidates

Rank	Candidate	Relevance	Business	Context	Overall	Confidence
1	Fast Pass	92	85	96	91	89%
2	Family Meal	88	72	94	84	86%
3	Annual Membership	81	90	76	83	82%
4	Guided Tour	76	65	82	74	79%
5	Souvenir Bundle	64	55	71	63	68%

Score Breakdown

Select:

Fast Pass

Show:

Current Basket Relevance — High

Party Relevance — High

Visit-Date Relevance — High

Availability — Available

Product Relationship — Strong

Business Priority — High

Important

Scores should be normalized/explained appropriately. The UI should not pretend every AI model literally produces these exact human-readable percentages internally.

These can be TICVAI's normalized decision scores.

Manual Business Pinning

Where governance allows:

Pin Candidate

should mean business priority/merchandising override.

It must be logged.

Actions

Inspect Score | Compare Candidate | Pin if Authorized | Continue

Screen 6 — Next-Best-Product & Next-Best-Action Decisioning — Continued

No-Sell Recommendation

TICVAI must support:

No Commercial Recommendation

For example, if a customer has just experienced a service issue, the best next action may be:

Customer Service Follow-Up

rather than trying to upsell another product.

Similarly:

Membership Expiring Soon

→ Recommend Renewal

Customer has incomplete reservation

→ Complete Reservation

Customer is approaching an attraction with a valid entitlement

→ Use Existing Entitlement

rather than:

→ Buy Another Ticket

Decision Workspace

Show:

Candidate Action	Type	Eligibility	Relevance	Priority	Result
Renew Membership	Action	Eligible	94	High	Selected
Book Next Visit	Action	Eligible	76	Medium	Alternate
Retail Bundle	Product	Eligible	58	Low	Not Selected
New Membership	Product	Ineligible	—	—	Excluded

Action Source

Each action should identify its owning TICVAI module:

Renew Membership → Membership

Join Virtual Queue → Virtual Queue

Upgrade Ticket → Ticketing

Use Loyalty Benefit → Loyalty

Reserve Experience → Ticketing / Reservation

Complete Payment → Commerce / Payment

Recommendation Intelligence selects the appropriate action but does not duplicate the execution functionality of the owning module.

Actions

Configure Action Types | Inspect Decision | Test Next-Best-Action | Continue

Screen 7 — Ranking Policy, Diversity & Recommendation Mix

Purpose

Control how TICVAI turns ranked candidates into the final recommendation set presented to Board 4.

Selecting the three highest raw scores is not always the best customer experience.

For example, TICVAI should avoid returning:

1. Meal Package A
2. Meal Package B
3. Meal Package C

when a more useful recommendation set might be:

1. Fast Pass
2. Family Meal
3. Annual Membership

Final Recommendation Policy

Configure:

Maximum Recommendations

Example: 3

Minimum Recommendations

Example: 0

Maximum per Product Category

Example: 1

Maximum Commercial Recommendations

Example: 3

Diversity Controls

Allow diversification by:

- Product Category
- Recommendation Type
- Price Band
- Experience Type
- Business Objective
- Product Family
- Journey Purpose

Example

Initial ranking:

Rank	Candidate	Category	Score
1	Family Meal A	F&B	94
2	Family Meal B	F&B	92
3	Fast Pass	Upgrade	90
4	Family Meal C	F&B	88
5	Annual Membership	Membership	86

Configured policy:

Maximum 1 F&B recommendation.

Final set:

#1 Family Meal A

#2 Fast Pass

#3 Annual Membership

Repetition Control

Avoid repeatedly recommending the same product where Board 1 suppression/frequency policies apply.

Example:

Customer rejected:

Annual Membership

twice during current journey.

Policy:

Suppress for current session

Business Mix

Optional configurable mix:

- Revenue Recommendation
- Experience Recommendation
- Retention Recommendation

But the business should not be forced to fill every category.

Recommendation Position

Board 3 can return:

Rank 1 / Rank 2 / Rank 3

but **Board 4 owns actual screen placement and presentation.**

Diversity Override

Where necessary:

A substantially stronger candidate may remain even if diversity would normally replace it.

This behavior must be policy-driven and explainable.

Actions

Configure Ranking Policy | Test Diversity | Preview Final Set | Save

Screen 8 — Confidence, Uncertainty & Decision Quality Control

Purpose

Prevent TICVAI from presenting weak AI decisions as strong personalized recommendations.

Every recommendation decision should have an appropriate quality/confidence assessment.

Decision Confidence

Example:

Fast Pass

Confidence:

89% — High

Family Meal

Confidence:

84% — High

Annual Membership

Confidence:

62% — Medium

Confidence Bands

Configurable:

High

≥ 80%

Medium

60–79%

Low

< 60%

The actual ranges should be configurable according to the selected model/decision approach.

Confidence Inputs

Confidence may consider:

- Context Completeness
- Signal Availability
- Candidate Separation
- Model Confidence
- Historical Evidence
- Cold-Start State
- Data Freshness
- Rule Certainty

Decision Quality Checks

Before returning a recommendation, validate:

- ✓ Candidate still eligible
- ✓ Product still available
- ✓ Required context valid
- ✓ Signal freshness acceptable
- ✓ No new suppression
- ✓ Confidence threshold passed
- ✓ Recommendation policy satisfied

Uncertainty Handling

Example:

Top candidates:

Fast Pass — 71

Guided Tour — 70

Family Meal — 69

The engine should recognize:

Low ranking separation

rather than falsely claiming the first candidate is clearly superior.

Low Confidence Behavior

Configurable:

Use AI Result

Use Rule-Based Ranking

Use Business Priority

Use Popular Eligible

Return Generic Recommendation

Return No Recommendation

Context Quality Impact

Example:

Known customer:

Context Completeness 94%

Confidence:

88%

Anonymous customer:

Context Completeness 52%

Confidence:

61%

TICVAI can still recommend, but the recommendation should not be represented as equally personalized.

Quality Status

- Strong
- Acceptable
- Limited
- Low Confidence
- Fallback Required
- Decision Rejected

Actions

[Configure Threshold](#) | [Test Confidence](#) | [View Quality Factors](#) | [Configure Fallback](#)

Screen 9 — Recommendation Explainability & Decision Trace

Purpose

Explain **why TICVAI selected a recommendation** and why other candidates were not selected.

This is essential for administration, support, debugging, governance, and future AI audit.

Decision Header

Decision ID

REC-DEC-008821

Strategy

Family Visit Upsell

Context

CTX-20260919-10082

Selected

Fast Pass

Decision Mode

Hybrid AI + Rules

Confidence

89%

Customer-Friendly Reason

Where appropriate:

Popular upgrade for your selected visit.

This wording can later be used by Board 4 if the channel supports recommendation explanations.

Administrative Explanation

Example:

Fast Pass ranked #1 because:

- ✓ Customer has valid admission
- ✓ Fast Pass is available
- ✓ Party context matches strategy
- ✓ Visit date is eligible
- ✓ Strong product relationship
- ✓ High basket relevance
- ✓ Business priority is High
- ✓ No suppression applies

Decision Factor View

Factor	Contribution	Source
Basket Relevance	Strong	Current Basket
Product Relationship	Strong	Board 1
Party Context	Strong	Board 2
Visit Context	Strong	Board 2
Business Priority	High	Board 1
Availability	Valid	Capacity

Why Not Candidate?

Select:

VIP Admission

Result:

Not ranked because the product was removed before AI scoring.

Reason:

Sold Out

Select:

Guided Tour

Result:

Eligible but ranked below Fast Pass because contextual relevance was lower.

Decision Trace

Visual flow:

Strategy

↓

Context

↓

48 Candidates

↓

10 Eligible

↓

AI Scoring

↓

Ranking

↓

Diversity Policy

↓

Confidence Validation

↓

Fast Pass Selected

Provenance

Show which information came from:

- Board 1 Strategy
- Board 2 Context
- AI Model
- Business Rule
- Operational System
- Recommendation Policy

Important Requirement

Explainability must not fabricate a simplistic explanation that is unsupported by the actual decision process.

If the underlying model cannot provide a precise causal explanation, TICVAI should present:

Decision Factors

rather than claiming:

“This exact factor caused the recommendation.”

Actions

[View Full Trace](#) | [Compare Candidates](#) | [Export Decision](#) | [View Context](#) | [View Strategy](#)

Screen 10 — Decision Simulation, Comparison & Validation Lab

Purpose

Allow administrators to safely test Recommendation Intelligence before activating it in customer-facing journeys.

This becomes the final output of Board 3.

Simulation Header

Strategy

Fast Pass & Experience Upsell

Decision Mode

Hybrid

Environment

Simulation

Scenario Builder

Administrators can construct:

Customer Type

Returning Visitor

Membership

None

Loyalty

Silver

Party

2 Adults + 2 Children

Channel

B2C

Journey

Checkout

Venue

Theme Park

Visit

Saturday

Basket

2 Adult + 2 Child Admission

Context Source

Choose:

- Synthetic Scenario
- Existing Test Customer
- Anonymous Context
- Historical Context Snapshot
- Saved Scenario

Production personal data usage must follow applicable permissions.

Candidate Pipeline

Show:

48 Potential

→ 42 Active

→ 35 Channel Eligible

→ 18 Context Eligible

→ 10 Final Candidates

→ 10 Scored

→ 3 Returned

Decision Result

#1 Fast Pass

Overall Decision Score: **91**

Confidence: **89%**

Reason:

Strong basket, visit and product relationship relevance

#2 Family Meal

Score: **84**

Confidence: **86%**

#3 Annual Membership

Score: **83**

Confidence: **82%**

Compare Decision Modes

This is important.

Administrators should be able to compare:

Candidate	AI	Rules	Hybrid	Business Priority
Fast Pass	#1	#2	#1	#2
Family Meal	#2	#3	#2	#3
Annual Membership	#3	#1	#3	#1

This helps the business understand what AI is changing.

Scenario Comparison

Run:

Family Visitor

vs.

Solo Visitor

vs.

Annual Member

vs.

Anonymous Visitor

The recommendation results should change according to context.

Sensitivity Test

Allow controlled changes such as:

What happens if Fast Pass capacity reaches zero?

Expected:

Fast Pass removed before ranking.

What happens if customer becomes Annual Member?

Expected:

Annual Membership acquisition recommendation suppressed.

Quality Checks

Show:

- Eligibility Passed
- Hard Rules Passed
- Context Valid
- Ranking Completed
- Diversity Passed
- Confidence Passed
- Explanation Available
- Fallback Available

Validation Issues

Classify:

Blocking

Warning

Recommendation

Actions

Run Simulation | Compare Modes | Save Scenario | View Decision Trace | Return to Configuration

Primary CTA

Approve Decision Strategy / Continue to Personalized Delivery →

This moves the approved decision output into **Board 4**.

Critical Board 3 Decision Architecture

Soft Labs should not implement this as:

Customer data → LLM → recommendation.

That would give TICVAI insufficient business control.

The architecture should be:

BOARD 1

Recommendation Strategy

+

BOARD 2

Recommendation Context



Candidate Generation



Hard Eligibility



Exclusions & Suppression



Operational Availability



Business Guardrails



Eligible Candidate Set



Feature / Signal Preparation



AI / Rules / Hybrid Decision Engine



Candidate Scoring



Ranking



Diversity / Mix Policy



Confidence & Quality Control



Explainability



FINAL RECOMMENDATION SET



BOARD 4

Channel & Journey Delivery

Recommendation Decision Object

Every runtime decision should create a structured object.

Conceptually:

RecommendationDecision

|

- └─ Decision ID
- └─ Timestamp
- └─ Tenant
- └─ Venue
- └─ Strategy ID
- └─ Context ID
- └─ Decision Mode
- |
- └─ Candidate Funnel
 - | └─ Original Candidates
 - | └─ Excluded Candidates
 - | └─ Exclusion Reasons
 - | └─ Eligible Candidates
- |
- └─ Scoring
 - | └─ Features Used
 - | └─ Model / Rule Set
 - | └─ Candidate Scores
 - | └─ Ranking
- |
- └─ Final Recommendations
 - | └─ Rank
 - | └─ Product / Action
 - | └─ Confidence
 - | └─ Explanation
- |
- └─ Fallback
- └─ Decision Latency
- └─ Model / Engine Version
- └─ Decision Trace

This object becomes the input for Board 4 and later the measurement basis for Board 5.

Hybrid Decisioning — Recommended TICVAI Default

For TICVAI, I recommend **Hybrid Decisioning** as the default rather than completely free AI ranking.

The hierarchy should be:

1. Hard Business Rules

Cannot be overridden.

↓

2. Eligibility / Availability / Consent

Cannot be overridden.

↓

3. Business Guardrails

Control what AI may do.

↓

4. AI Intelligence

Determines relevance among permitted options.

↓

5. Ranking Policy

Controls final mix.

↓

6. Confidence

Determines whether the AI result is strong enough.

↓

7. Fallback

Used when AI is unavailable or insufficiently confident.

This gives TICVAI the intelligence of AI without surrendering commercial or operational control.

Decision Engine vs LLM

Another important point for Soft Labs:

The Recommendation Engine should **not depend exclusively on a generative LLM**.

TICVAI should be able to use different decision technologies:

- Rules
- Statistical Models
- Machine Learning Ranking
- Recommendation Models
- Embeddings / Similarity
- Collaborative Signals
- Contextual Models
- AI Services
- LLM reasoning where appropriate
- Hybrid combinations

The recommendation orchestration layer should abstract the provider/model.

That will become particularly important later when we design the **AI Platform, Models & Providers** area.

Model Independence

Board 3 should request something conceptually like:

Rank these 10 eligible candidates for this Recommendation Context under Strategy X.

It should not care whether the underlying intelligence is ultimately:

TICVAI Model

Azure/OpenAI

AWS

Google

Customer-Hosted Model

or another approved provider.

The AI platform layer should handle that abstraction.

Real-Time Decision Requirement

Recommendation decisioning will often occur inside:

B2C

Mobile App

POS

Kiosk

Therefore latency matters.

The engine should support:

- Configurable decision timeout
- Cached/configured fallback
- Rule-based fallback
- Circuit breaker
- Model availability monitoring

A slow AI provider should **not block checkout**.

Example:

AI timeout after configured threshold.

↓

Immediately use:

Rule-Based Recommendation

rather than delay payment.

Revalidation Before Delivery

A recommendation may be valid at **14:20:01** and invalid at **14:20:20**.

Example:

Last Fast Pass sold.

Therefore TICVAI must support final validation close to execution/display where appropriate.

Board 3 decides:

Fast Pass is #1.

Before purchase, the owning commerce/product systems must still confirm:

Fast Pass remains saleable.

AI ranking is never a replacement for transactional validation.

Cold-Start Strategy

Board 3 must support:

New Customer

Use:

Current Context + Basket + Product Relationships + Business Rules

Anonymous Visitor

Use:

Session + Journey + Basket + Visit Context

New Product

Use:

Product Metadata + Relationships + Business Strategy

New Venue

Use:

Templates + Business Rules + Contextual Intelligence

The recommendation engine should therefore remain useful before TICVAI has accumulated years of behavioral data.

Feedback Boundary

Board 3 can **consume learned/model information**, but the detailed feedback and optimization loop belongs to Board 5.

For example:

Customer sees Fast Pass.

↓

Customer accepts.

That outcome becomes performance feedback.

But Board 3 should not contain the entire:

Attribution → Experiment → Uplift → Optimization

system.

That belongs to **Board 5**.

Important Boundary — Board 4

Board 3 determines:

WHAT should be recommended?

and:

In what ranking?

Board 4 determines:

WHERE should it appear?

WHEN should it appear?

How should it be presented?

Which channel/placement receives it?

For example:

Board 3:

#1 Fast Pass

Board 4:

Display Fast Pass as an upsell card on B2C checkout after ticket selection.

That separation is important.

Important Boundary — Pricing

Board 3 can use:

Current Price

Margin where permitted

Commercial Value

as decision signals.

It must not calculate a new ticket price.

Pricing remains owned by TICVAI's **Pricing / Dynamic Pricing** modules.

Important Boundary — Promotions

Board 3 may recommend:

Family Meal

and an existing applicable promotion may make the offer:

AED 120 → AED 99

But Board 3 does not create that discount.

Promotion configuration remains with **Promotions**.

Board 3 — Final Screen Map

Recommendation Decisioning, Ranking & AI Intelligence

- 1. Recommendation Decisioning Command Center**
- 2. Decision Strategy & Mode Configuration**
- 3. Candidate Funnel & Pre-Ranking Engine**
- 4. AI Scoring & Feature Configuration**
- 5. Candidate Scoring & Ranking Workspace**
- 6. Next-Best-Product & Next-Best-Action Decisioning**
- 7. Ranking Policy, Diversity & Recommendation Mix**
- 8. Confidence, Uncertainty & Decision Quality Control**
- 9. Recommendation Explainability & Decision Trace**
- 10. Decision Simulation, Comparison & Validation Lab**

This completes the actual **intelligence brain** of the recommendation system.

The sequence is now very clean:

Board 1 — What are we allowed and trying to recommend?

→ **Board 2 — Who is the customer and what is happening?**

→ **Board 3 — What is the best recommendation/action?**

→ **Board 4 — Where, when and how do we deliver it?**

→ **Board 5 — Did it work, and how do we improve it?**

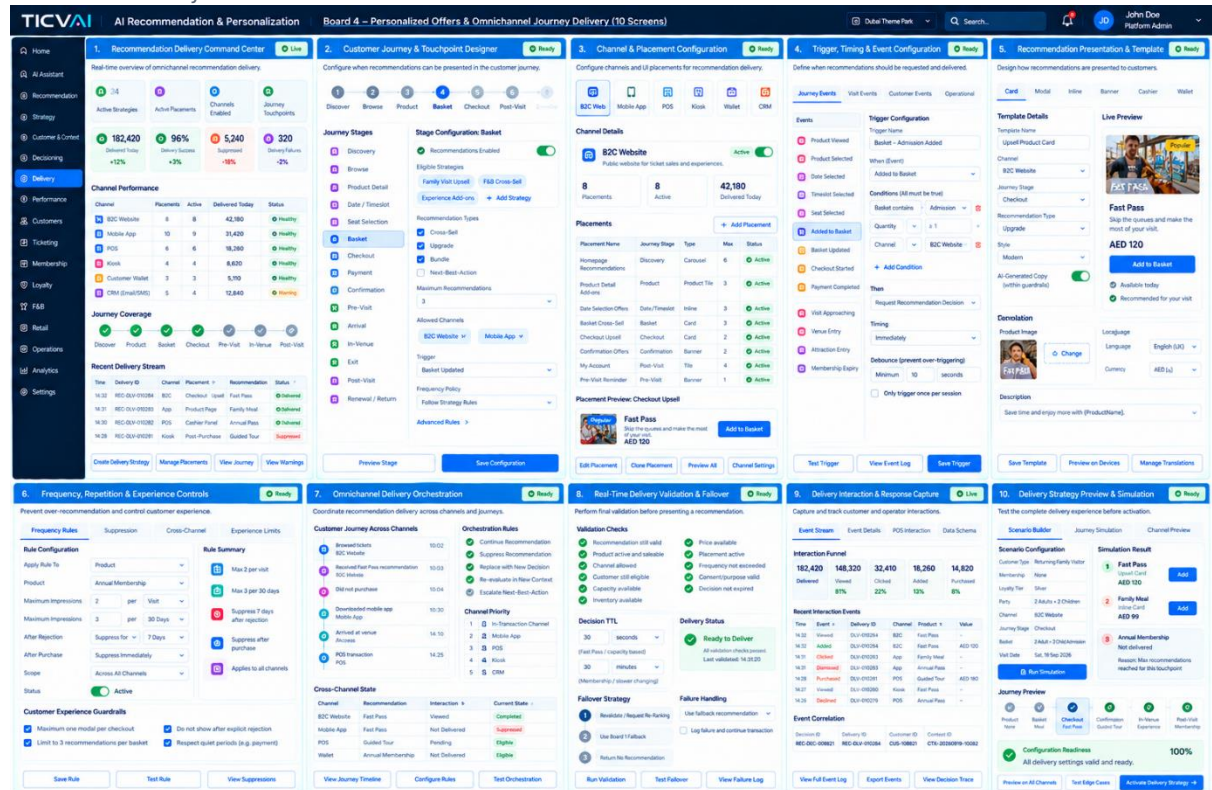
Recommendation & Personalization — Board 4: Personalized Offers & Omnichannel Journey Delivery

Module: AI Recommendation & Personalization

Board: 4 of 5

Screens: 10

Board Type: Recommendation Delivery / Journey Orchestration / Placement / Channel Configuration / Frequency / Real-Time Delivery



Board Purpose

Boards 1–3 have already determined:

Board 1: What are we trying and allowed to recommend?

Board 2: Who is the customer and what is their current context?

Board 3: What is the best recommendation or next-best-action?

Board 4 answers the next critical question:

Where, when, and how should TICVAI deliver that recommendation to the customer or frontline operator?

For example, Board 3 may return:

- #1 Fast Pass Upgrade
- #2 Family Meal Package
- #3 Annual Membership

Board 4 determines whether those recommendations appear:

- On the B2C product page
- During checkout
- On the mobile app

- At POS
- On a kiosk
- In the customer wallet
- Pre-visit
- In-venue
- Post-visit
- Through CRM-triggered journeys where applicable

The architecture is:

Board 3 Recommendation Decision

↓

Journey Eligibility

↓

Channel Selection

↓

Placement Selection

↓

Timing / Trigger

↓

Frequency & Suppression

↓

Presentation Template

↓

Real-Time Revalidation

↓

Delivery

↓

Interaction Event

↓

Board 5 — Performance & Optimization

Board 4 does **not decide what is relevant**. Board 3 does that. Board 4 controls the delivery experience.

Screen 1 — Recommendation Delivery Command Center

Purpose

Provide a centralized operational view of all recommendation delivery configurations across TICVAI channels and customer journeys.

Header KPIs

- Active Delivery Strategies
- Active Placements
- Channels Enabled
- Journey Touchpoints
- Recommendations Delivered Today
- Delivery Success Rate
- Suppressed Deliveries
- Expired Decisions
- Delivery Failures
- Configuration Warnings

Channel Overview

Show:

Channel	Placements	Active	Delivered Today	Status
B2C Web	8	8	42,180	Healthy
Mobile App	10	9	31,420	Healthy
POS	6	6	18,280	Healthy
Kiosk	4	4	8,620	Healthy
Wallet	3	3	5,110	Healthy
CRM	5	4	12,840	Warning

Journey Coverage

Visual journey:

Discovery → Product → Basket → Checkout → Confirmation → Pre-Visit → Arrival → In-Venue → Exit → Post-Visit

Show which stages have recommendation placements.

Delivery Health

Statuses:

- Healthy
- Placement Missing
- Channel Unavailable
- High Suppression

- High Expiry
- Delivery Error
- No Eligible Recommendation

Recent Delivery Stream

Show:

- Delivery ID
- Decision ID
- Channel
- Journey Stage
- Placement
- Recommendation
- Status
- Timestamp

Actions

[Create Delivery Strategy](#) | [Configure Channel](#) | [Manage Placements](#) | [View Journey](#) | [View Warnings](#)

Screen 2 — Customer Journey & Touchpoint Designer

Purpose

Define exactly **at which point in the customer journey** recommendations may be presented.

This should be a visual journey configuration screen.

Journey Map

Display:

Discovery

↓

Browse

↓

Product Detail

↓

Date / Timeslot

↓

Seat Selection

↓

Basket

↓

Checkout



Payment



Confirmation



Pre-Visit



Arrival



In-Venue



Exit



Post-Visit



Renewal / Return

Touchpoint Configuration

Select a stage.

Example:

Basket

Configure:

- Recommendations Enabled
- Eligible Strategies
- Eligible Recommendation Types
- Maximum Recommendations
- Allowed Channels
- Placement
- Trigger
- Frequency Policy

Example

Journey Stage

Basket

Strategy

Family Visit Upsell

Recommendation Types

Cross-Sell + Upgrade

Maximum Recommendations

3

Channels

B2C + Mobile App

Journey-Specific Rules

Example:

At **Product Detail**:

Show complementary experiences.

At **Basket**:

Show add-ons and upgrades.

At **Checkout**:

Limit recommendation interruption.

At **Confirmation**:

Recommend pre-bookable experiences.

At **In-Venue**:

Prioritize time-sensitive experiences.

At **Post-Visit**:

Prioritize membership / return visit.

Journey Conflict

If multiple strategies target the same stage:

TICVAI should apply:

- Placement priority
- Strategy priority
- Board 3 ranking
- Frequency rules

Actions

Add Touchpoint | Configure Stage | Disable Stage | Preview Journey | Save

Screen 3 — Channel & Placement Configuration

Purpose

Configure the channels and exact UI/business placements where recommendation decisions can be delivered.

Supported Channels

Include:

- B2C Web
- Mobile App
- POS
- Mobile / Flying POS
- Kiosk
- Customer Wallet
- Agent / Customer Service
- B2B Portal where applicable
- CRM-triggered journey
- Other supported digital channels

Channel Configuration

Example:

B2C Web

Status:

Active

Placements:

- Homepage Recommendation
- Product Detail
- Product Selection
- Basket
- Checkout
- Confirmation
- My Account
- Pre-Visit

Placement Object

Each placement should contain:

- Placement ID
- Placement Name
- Channel
- Journey Stage
- Placement Type
- Maximum Items
- Allowed Recommendation Types
- Strategy Eligibility
- Priority
- Status

Example

Placement

B2C-CHECKOUT-UPSELL

Channel

B2C

Journey

Checkout

Type

Recommendation Card

Maximum Items

2

POS Placement

Example:

Cashier Recommendation Panel

This is different from B2C because the recommendation is primarily shown to the cashier.

The UI may say:

Recommended Upgrade: Annual Pass

with:

Why: Customer qualifies and upgrade value is favorable.

Cashier can:

Offer | Skip | View Details

Kiosk Placement

Must consider:

- Limited screen area
- Transaction speed
- Touch interface
- Maximum recommendation count

Actions

Create Placement | Configure | Clone Placement | Disable | Preview

Screen 4 — Trigger, Timing & Event Configuration

Purpose

Define **when** TICVAI should request or display a recommendation.

Recommendations should be event-driven rather than displayed randomly.

Trigger Types

Support:

Journey Event

- Product Viewed
- Product Selected
- Date Selected
- Timeslot Selected
- Seat Selected
- Added to Basket
- Basket Updated
- Checkout Started
- Payment Completed

Visit Event

- Visit Approaching
- Ticket Activated
- Venue Entry
- Attraction Entry
- Attraction Exit
- Zone Change where permitted
- Visit Ending
- Venue Exit

Customer Event

- Membership Near Expiry
- Loyalty Milestone
- Return Visit
- Account Login

Operational Event

- Capacity Available
- Timeslot Availability
- Experience Starting Soon
- Product Becomes Available

Trigger Builder

Example:

WHEN

Customer enters Basket

AND

Basket contains Admission

AND

Recommendation Strategy is eligible

THEN

Request Recommendation Decision

Timing

Configure:

- Immediately
- After X seconds
- X minutes before event
- X hours before visit
- X days before visit
- After completion
- Scheduled Window

Example

Guided Tour Recommendation

Trigger:

Customer has entered venue

AND

next eligible Guided Tour starts within 90 minutes.

Trigger Debounce

Prevent excessive decision calls.

Example:

Do not request a new recommendation every time basket quantity changes within 2 seconds.

Actions

Create Trigger | Test Trigger | Configure Timing | View Events

Screen 5 — Recommendation Presentation & Experience Template

Purpose

Configure how the recommendation should be presented without allowing the AI decision engine to arbitrarily construct customer-facing experiences.

Presentation Types

Support:

- Recommendation Card
- Product Tile
- Upsell Modal
- Inline Recommendation

- Basket Add-On
- Checkout Upsell
- Carousel
- Banner
- Cashier Prompt
- Kiosk Prompt
- Wallet Card
- Notification Content Reference
- Next-Best-Action Card

Template Configuration

Example:

Fast Pass Upsell Card

Fields:

- Product Image
- Product Name
- Short Description
- Current Price
- Promotional Price where applicable
- Recommendation Reason
- CTA
- Availability
- Badge

CTA Options

Examples:

- Add
- Upgrade
- View
- Reserve
- Join
- Renew
- Add to Basket
- Learn More

AI-Generated Copy

If TICVAI allows AI-generated presentation text, it should be separately governed.

Options:

Approved Template Only

AI Fill Approved Fields

AI Rewrite Within Guardrails

Example

Approved template:

Upgrade your visit with **{Product Name}** for **{Price}**.

AI should not independently invent:

“Only 2 left!”

unless inventory actually supports that claim and such messaging is permitted.

Localization

Support:

- Language
- Locale
- Currency formatting
- RTL
- Channel-specific copy

Preview

Preview:

Desktop

Mobile

POS

Kiosk

Actions

Create Template | Edit | Preview | Localize | Save

Screen 6 — Frequency, Repetition & Customer Experience Controls

Purpose

Prevent over-recommendation and protect the customer experience across channels.

This is essential because TICVAI could otherwise recommend the same product repeatedly on B2C, app, POS and in-venue.

Frequency Controls

Configure by:

- Recommendation
- Product

- Strategy
- Placement
- Channel
- Journey
- Customer
- Session

Example

Annual Membership Recommendation

Maximum:

2 impressions per visit

3 impressions per 30 days

After rejection:

Suppress for 7 days

After purchase:

Suppress immediately

Cross-Channel Frequency

Example:

Customer sees Annual Membership on:

B2C Checkout

Then arrives at:

POS

TICVAI should know that the customer has already received the recommendation if identity/context permits.

Frequency Policies

Support:

- Per Session
- Per Transaction
- Per Visit
- Per Day
- Per Week
- Per Month
- Custom Window

Rejection Handling

Customer action:

Not Interested

Configure:

- Suppress Current Session
- Suppress Current Visit
- Suppress X Days
- Suppress Strategy
- Reduce Recommendation Priority

No-Response Handling

Do not automatically treat:

No Click

as:

Customer Rejected

These are different interaction states.

Customer Experience Guardrails

Examples:

Maximum one interruption/modal per checkout.

Maximum three recommendation cards per basket.

Do not show another upsell immediately after explicit rejection.

Do not interrupt payment processing.

Actions

Create Frequency Policy | Configure Suppression | Test Cross-Channel | Save

Screen 7 — Omnichannel Delivery Orchestration

Purpose

Coordinate recommendation delivery when the same customer journey exists across multiple channels.

This is more than enabling channels independently.

Journey Example

Customer:

Browses B2C

↓

Adds Admission

↓

Receives Fast Pass recommendation

↓

Does not purchase

↓

Downloads App

↓

Arrives at Venue

↓

Uses POS

The system must decide whether Fast Pass should:

- Be shown again
- Be suppressed
- Be replaced
- Be reprioritized
- Be presented differently

Channel Priority

Configure:

1. In-Transaction Channel
2. Customer App
3. POS
4. Kiosk
5. CRM

depending on strategy.

Continuity Rules

Support:

Continue Recommendation

Suppress Recommendation

Replace Recommendation

Re-Evaluate

Escalate Next-Best-Action

Example

B2C:

Fast Pass offered but ignored.

In-Venue:

Capacity is now limited.

Board 3 re-evaluates.

New result:

Guided Tour becomes #1.

Board 4 should deliver the **new decision**, not blindly repeat the stale B2C recommendation.

Cross-Channel State

Show:

Channel	Recommendation	Interaction	Current State
B2C	Fast Pass	Viewed	Completed
App	Fast Pass	Not Delivered	Suppressed
POS	Guided Tour	Pending	Eligible

Anonymous → Known Continuity

If an anonymous session later authenticates:

TICVAI may connect recommendation context where permitted.

Do not assume identity without a valid resolution.

Actions

[Configure Orchestration](#) | [View Journey](#) | [Set Channel Priority](#) | [Test Continuity](#)

Screen 8 — Real-Time Delivery Validation & Failover

Purpose

Perform the final safety check immediately before a recommendation is presented.

A Board 3 recommendation is a decision, not a guarantee that the product remains available.

Final Validation

Before display:

- ✓ Recommendation still valid
- ✓ Product active
- ✓ Product saleable
- ✓ Channel allowed
- ✓ Customer still eligible
- ✓ Capacity available
- ✓ Inventory available
- ✓ Price available
- ✓ Placement active
- ✓ Frequency not exceeded
- ✓ Consent/purpose valid
- ✓ Decision not expired

Example

Board 3 at:

14:20:01

recommended:

Fast Pass

At:

14:20:08

Fast Pass inventory becomes:

0

Board 4 final validation:

Do Not Deliver

Failover

Configured behavior:

Request Re-Ranking

↓

If unavailable:

Use Board 1 Fallback

↓

If unavailable:

No Recommendation

Decision TTL

Recommendation decisions should support a configurable:

Time To Live

Example:

Fast-changing capacity recommendation:

TTL = 30 seconds

Membership recommendation:

TTL = 30 minutes

Service Failure

If AI decision service unavailable:

Board 4 should be capable of consuming the approved fallback output.

Checkout should not fail because recommendations are unavailable.

Delivery Error Categories

- Decision Expired
- Product Unavailable
- Capacity Changed
- Price Changed
- Placement Failure
- Channel Failure
- Validation Failure
- Frequency Suppression

- Consent Restriction
- Recommendation Service Unavailable

Actions

[Configure TTL](#) | [Test Failover](#) | [Revalidate](#) | [View Failure Log](#)

Screen 9 — Delivery Interaction & Response Capture

Purpose

Capture what happened after TICVAI delivered the recommendation.

This creates the event stream Board 5 will use for performance, attribution and optimization.

Interaction Lifecycle

A recommendation can become:

Requested

↓

Generated

↓

Eligible for Delivery

↓

Delivered

↓

Viewed / Presented

↓

Customer/Operator Action

Interaction Events

Support:

- Delivered
- Impression
- Viewed
- Clicked
- Expanded
- Added
- Accepted
- Declined
- Dismissed
- Ignored
- Removed

- Purchased
- Action Completed
- Expired
- Suppressed
- Delivery Failed

Important Distinction

These must not be treated as identical:

Delivered

≠

Viewed

≠

Clicked

≠

Accepted

≠

Purchased

Example

Recommendation

Fast Pass

Decision ID

REC-DEC-008821

Delivery ID

REC-DLV-010284

Channel

B2C

Placement

Checkout Upsell

Delivered

14:20:05

Viewed

14:20:07

Added

14:20:14

Purchased

14:22:42

POS Interaction

For cashier-assisted recommendations:

Track separately:

Recommendation shown to cashier

Cashier offered recommendation

Customer accepted

This is important.

Otherwise TICVAI might incorrectly conclude a recommendation failed when the cashier never presented it.

Event Correlation

Every event should retain:

- Decision ID
- Delivery ID
- Strategy ID
- Context ID
- Customer/session reference
- Placement
- Channel
- Product/action
- Timestamp

Actions

View Interaction | Inspect Journey | Export Events | View Decision Trace

Screen 10 — Delivery Strategy Preview & Journey Simulation

Purpose

Validate the entire Board 4 delivery configuration before activation.

This should simulate:

Decision → Channel → Placement → Timing → Validation → Customer Interaction

Scenario

Customer

Returning Family Visitor

Channel

B2C

Journey

Checkout

Basket

2 Adult + 2 Child

Board 3 Result:

1. Fast Pass
2. Family Meal
3. Annual Membership

Delivery Configuration

Checkout Placement

Maximum Recommendations:

2

Allowed:

- Upgrade
- Cross-Sell

Modal:

Not Allowed

Inline:

Allowed

Final Delivery

Therefore:

Position 1

Fast Pass

Presentation:

Checkout Upsell Card

CTA:

Add Fast Pass

Position 2

Family Meal

Presentation:

Compact Add-On Card

CTA:

Add Meal Package

Annual Membership:

Not delivered at this touchpoint

Reason:

Maximum recommendation count reached.

Journey Simulation

Show the customer progressing:

Product

→ No Recommendation

Basket

→ Family Meal

Checkout

→ Fast Pass

Confirmation

→ Guided Tour

Pre-Visit

→ Experience Reservation

In-Venue

→ Contextual Recommendation

Post-Visit

→ Membership

This does not mean every stage must have a recommendation. It shows configured possibilities.

Channel Comparison

Simulate the same Board 3 decision across:

Channel	Delivery
B2C	Visual Recommendation Card
Mobile	Compact Card
POS	Cashier Recommendation
Kiosk	Single Upsell
Wallet	Contextual Card

Validation Results

Show:

Journey Configuration — Passed

Placement Configuration — Passed

Frequency Policy — Passed

Final Validation — Passed

Fallback — Configured

Event Tracking — Configured

Configuration Readiness

100% — Ready

Actions

Run Simulation | Change Channel | Preview Journey | Test Failover | Save Draft

Primary CTA

Activate Delivery Strategy / Continue to Performance & Optimization →

Critical Board 4 Architecture

I recommend Soft Labs implement Board 4 as a **Recommendation Delivery Orchestrator**, not embed recommendation logic independently inside B2C, POS, app and kiosk.

The architecture should be:

Board 3

Recommendation Decision

↓

Delivery Orchestrator

↓

Journey Eligibility

↓

Trigger / Timing

↓

Channel

↓

Placement

↓

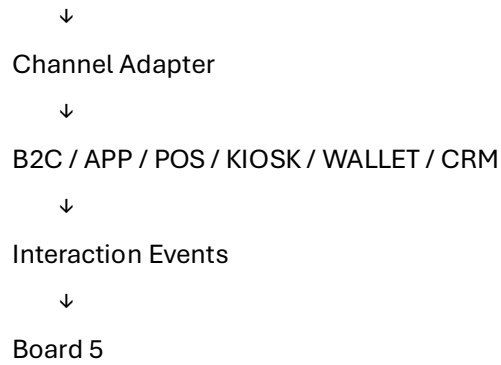
Frequency / Suppression

↓

Final Real-Time Validation

↓

Presentation Template



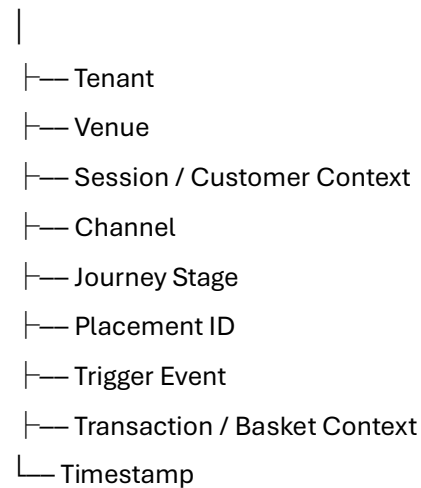
Performance & Optimization

This gives TICVAI **one central recommendation intelligence layer** while allowing every channel to present recommendations appropriately.

Delivery Request Object

When a channel requests a recommendation, I recommend a structured request such as:

RecommendationDeliveryRequest

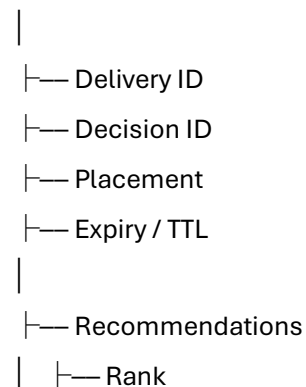


The system can then resolve the appropriate strategy/context/decision.

Delivery Response Object

Board 4 should return a channel-safe response:

RecommendationDeliveryResponse



- | |— Product / Action ID
- | |— Presentation Template
- | |— Display Content
- | |— Price Reference
- | |— CTA
- | |— Explanation Reference
- |
- |— Tracking References

The channel should not need to understand how AI generated the recommendation.

Important — POS Recommendation Experience

For TICVAI, I would make POS recommendation delivery particularly strong because the **cashier and customer are different actors**.

Example:

Cashier sees

AI RECOMMENDATION

Annual Pass Upgrade

Customer can upgrade today's tickets for AED 320 additional.

Why:

Returning visitor + family purchase + eligible upgrade

Buttons:

Offer to Customer

Skip

Details

After cashier clicks:

Offer to Customer

the system records:

Cashier Presented Recommendation

If customer accepts:

Customer Accepted

This gives TICVAI much better recommendation analytics than simply knowing the recommendation appeared on the cashier's monitor.

Important — Recommendations Must Not Block Core Sales

This should be an explicit non-functional requirement.

If the Recommendation Engine or Delivery Orchestrator fails:

Ticket Sale must continue.

If recommendation response exceeds configured timeout:

Continue transaction without recommendation or use safe fallback.

AI personalization is an enhancement to commerce—not a dependency that can stop admission sales.

Important Boundary — CRM

Board 4 can trigger or provide recommendation content to CRM where appropriate.

For example:

Post-Visit

Board 3:

Annual Membership Upgrade

Board 4:

Eligible for post-visit CRM touchpoint.

But CRM remains responsible for its communication execution, campaign permissions and communication channel governance.

Recommendation should not rebuild the email/SMS/WhatsApp campaign engine.

Important Boundary — Channel UI

Board 4 defines:

Placement + Template + Content Structure + Recommendation

but each owning channel still owns its transactional UX.

For example:

B2C owns checkout.

POS owns cashier transaction flow.

Kiosk owns kiosk transaction flow.

Recommendation should plug into those experiences through defined placement components/APIs.

Important Boundary — Promotions & Pricing

Board 4 displays the **valid current commercial information** supplied by the owning modules.

It does not calculate:

- Price
- Discount
- Tax
- Fee
- Promotion eligibility

If Board 4 displays:

Family Meal — AED 99

the AED 99 must come from the applicable TICVAI pricing/promotion calculation.

Important Boundary — Board 5

Board 4 records:

What was delivered and what did the customer/operator do?

Board 5 answers:

Did the recommendation actually improve performance?

Therefore Board 4 captures events, while Board 5 handles:

- Conversion
 - Incremental Revenue
 - Attach Rate
 - AOV Uplift
 - Acceptance Rate
 - Attribution
 - A/B Testing
 - Control Groups
 - Strategy Comparison
 - Model Comparison
 - Recommendation Optimization
-

Board 4 — Final Screen Map

Personalized Offers & Omnichannel Journey Delivery

- 1. Recommendation Delivery Command Center**
- 2. Customer Journey & Touchpoint Designer**
- 3. Channel & Placement Configuration**
- 4. Trigger, Timing & Event Configuration**
- 5. Recommendation Presentation & Experience Template**
- 6. Frequency, Repetition & Customer Experience Controls**
- 7. Omnichannel Delivery Orchestration**
- 8. Real-Time Delivery Validation & Failover**
- 9. Delivery Interaction & Response Capture**
- 10. Delivery Strategy Preview & Journey Simulation**

This creates a clean separation across the first four boards:

- Board 1 — WHAT are we allowed and trying to recommend?**
- **Board 2 — WHO is the customer and WHAT is happening?**
 - **Board 3 — WHICH recommendation/action is best?**
 - **Board 4 — WHERE, WHEN and HOW should we deliver it?**
 - **Board 5 — DID IT WORK and how should TICVAI improve it?**

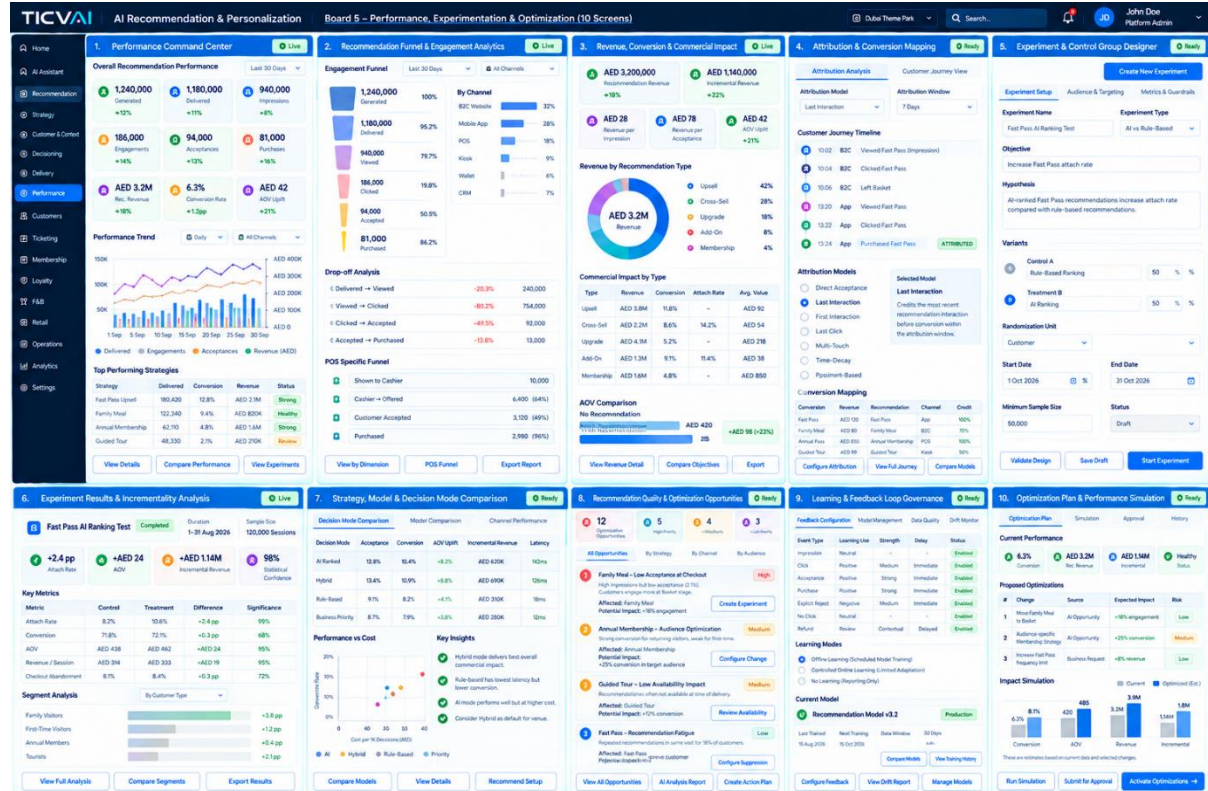
Recommendation & Personalization — Board 5: Recommendation Performance, Experimentation & Optimization

Module: AI Recommendation & Personalization

Board: 5 of 5

Screens: 10

Board Type: Performance Analytics / Attribution / Incrementality / Experimentation / Optimization / AI Feedback Loop



Board Purpose

Board 5 closes the recommendation lifecycle.

The first four boards answer:

Board 1: What are we trying and allowed to recommend?

Board 2: Who is the customer and what is happening?

Board 3: Which recommendation or action is best?

Board 4: Where, when and how should it be delivered?

Board 5 answers:

Did the recommendation actually work, what business impact did it create, and how should TICVAI improve future recommendation decisions?

The end-to-end loop becomes:

Strategy

→ Context

→ Decision

→ Delivery

- **Customer Response**
- **Conversion**
- **Attribution**
- **Incremental Impact**
- **Experimentation**
- **Optimization**
- **Controlled Feedback to Recommendation Intelligence**

A critical principle is that TICVAI must distinguish:

“The customer purchased after seeing a recommendation”

from:

“The customer purchased because of the recommendation.”

This board therefore needs both normal attribution and controlled experimentation/incrementality measurement.

Screen 1 — Recommendation Performance Command Center

Purpose

Provide management with one central view of recommendation performance across venues, strategies, channels, placements, products and AI decision modes.

Header KPIs

- Recommendations Generated
- Recommendations Delivered
- Impressions
- Engagements
- Acceptances
- Purchases / Completed Actions
- Recommendation Revenue
- Acceptance Rate
- Conversion Rate
- Attach Rate
- AOV Uplift
- Incremental Revenue
- Active Experiments

Performance Funnel

Example:

1,240,000 Decisions

↓

1,180,000 Delivered

↓

940,000 Viewed

↓

186,000 Engaged

↓

94,000 Accepted

↓

81,000 Purchased

Performance Trend

Chart by:

- Hour
- Day
- Week
- Month

Metrics:

- Delivery
- Conversion
- Revenue
- Incremental Revenue
- Acceptance
- AOV

Strategy Performance

Strategy	Delivered	Conversion	Revenue	Incremental Revenue	Status
Fast Pass Upsell	180K	12.8%	AED 2.1M	AED 610K	Strong
Family Meal	122K	9.4%	AED 820K	AED 240K	Healthy
Annual Pass	62K	4.8%	AED 1.6M	AED 380K	Strong
Guided Tour	48K	2.1%	AED 210K	AED 32K	Review

Filters

- Tenant
- Venue
- Strategy
- Use Case
- Channel
- Placement

- Journey Stage
- Product
- Recommendation Type
- Audience
- Decision Mode
- Experiment
- Date Range

Actions

View Strategy | Compare Performance | View Experiments | Analyze Revenue | View Optimization Opportunities

Screen 2 — Recommendation Funnel & Engagement Analytics

Purpose

Analyze exactly where customers interact with or abandon the recommendation journey. Board 4 captured the events; Board 5 analyzes them.

Funnel Stages

Track separately:

Generated

Eligible for Delivery

Delivered

Impression / Presented

Viewed

Clicked / Expanded

Added / Accepted

Purchased / Action Completed

Example

Stage	Count	Stage Rate
Generated	100,000	100%
Delivered	96,400	96.4%
Viewed	81,200	84.2%
Clicked	18,600	22.9%
Accepted	11,400	61.3%

Stage	Count	Stage Rate
Purchased	9,820	86.1%

Breakdown Dimensions

Analyze funnel by:

- Channel
- Placement
- Journey Stage
- Strategy
- Product
- Audience
- Customer Type
- Known vs Anonymous
- Device
- Venue
- Time
- Decision Mode

POS-Specific Funnel

Because we designed Board 4 to distinguish cashier and customer behavior, POS should show:

Recommendation Generated

↓

Displayed to Cashier

↓

Cashier Offered

↓

Customer Accepted

↓

Purchased

Example:

10,000 recommendations shown to cashiers.

Only 6,400 offered.

This tells management that the issue may be **cashier adoption**, not AI quality.

Drop-Off Analysis

Example:

Checkout Fast Pass

High views

but

Low add-to-basket.

Potential areas to investigate:

- Price
- Presentation
- Relevance
- Placement
- Timing

The system should identify correlations but not automatically claim causation.

Actions

Compare Funnel | Drill Down | View POS Funnel | Inspect Placement

Screen 3 — Revenue, Conversion & Commercial Impact Analytics

Purpose

Measure the commercial impact associated with Recommendation Intelligence.

Revenue KPIs

- Recommendation Revenue
- Incremental Revenue
- Revenue per Recommendation
- Revenue per Impression
- Revenue per Acceptance
- AOV
- AOV Uplift
- Attach Revenue
- Upgrade Revenue
- Cross-Sell Revenue
- Membership Revenue
- F&B Recommendation Revenue
- Retail Recommendation Revenue

Example

Customers Without Accepted Recommendation

Average Basket:

AED 420

Customers With Accepted Recommendation

Average Basket:

AED 518

Observed Difference:

+AED 98

But TICVAI should not automatically label the entire AED 98 as AI-generated incremental revenue. Controlled measurement from Screen 5/6 is needed for stronger causal claims.

Commercial Impact by Type

Type	Revenue	Conversion	Attach Rate	Avg Added Value
Upsell	AED 3.8M	11.8%	—	AED 92
Cross-Sell	AED 2.2M	8.6%	14.2%	AED 54
Upgrade	AED 4.1M	5.2%	—	AED 218
Add-On	AED 1.3M	9.1%	11.4%	AED 38

Objective Performance

Compare performance against Board 1 objective.
Example:

Strategy Objective

Increase Membership Conversion

Baseline

2.8%

Current

4.1%

Observed Uplift

+46%

If backed by controlled experiment:

Measured Incremental Uplift

+31%

Profit / Margin

Where valid margin data exists, allow:

- Incremental Margin
- Margin per Recommendation
- Margin Uplift

Do not estimate margin if the Finance/Pricing source does not provide it.

Actions

[View Revenue Detail](#) | [Compare Objectives](#) | [View Incrementality](#) | [Export Analysis](#)

Screen 4 — Recommendation Attribution & Conversion Mapping

Purpose

Determine which recommendation interactions should receive credit for a purchase or completed action.

This screen must prevent simplistic attribution.

Conversion Mapping

Example:

Customer journey:

10:02 Fast Pass shown on B2C

↓

10:04 Customer clicks Fast Pass

↓

10:06 Customer leaves basket

↓

13:20 Customer opens mobile app

↓

13:22 Fast Pass shown again

↓

13:24 Customer purchases Fast Pass

TICVAI needs a defined attribution policy.

Attribution Models

Support configurable models such as:

- Direct Acceptance
- Last Recommendation Interaction
- First Recommendation Interaction
- Last Click
- Multi-Touch
- Position-Based
- Time-Decay
- Experiment-Based Incrementality

Attribution Window

Configure by strategy/product:

- Same Session
- Same Transaction
- Same Visit
- X Hours
- X Days
- Custom

Example

Membership Upgrade

Attribution window:

7 days

In-Venue F&B

Attribution window:

2 hours

Conversion Types

Support:

- Product Purchase
- Upgrade Completed
- Membership Purchased
- Renewal Completed
- Add-On Added
- Reservation Completed
- Next-Best-Action Completed

Attribution Detail

Conversion	Revenue	Recommendation	Channel	Model	Credit
Fast Pass	AED 120	Fast Pass	App	Last Interaction	100%
Membership	AED 850	Annual Pass	B2C	Multi-Touch	70%

Important

Attributed Revenue and Incremental Revenue must remain separate.

Attributed Revenue

= revenue associated according to attribution rules.

Incremental Revenue

= estimated/measured additional revenue caused by recommendation intervention.

Actions

[Configure Attribution](#) | [View Journey](#) | [Compare Models](#) | [Inspect Conversion](#)

Screen 5 — Experiment & Control Group Designer

Purpose

Allow TICVAI to scientifically test recommendation strategies rather than assuming that higher observed conversion means the AI caused the improvement.

Experiment Types

Support:

- Recommendation vs No Recommendation
- AI vs Rule-Based
- AI vs Business Priority
- Strategy A vs Strategy B
- Placement A vs Placement B
- Presentation A vs Presentation B
- Ranking Model A vs Model B
- Recommendation Count Test
- Timing Test
- Channel Test

Experiment Builder

Example:

Experiment

Fast Pass AI Recommendation Test

Hypothesis

AI-ranked Fast Pass recommendations increase attach rate compared with rule-based recommendations.

Variants

Control A

Rule-Based Ranking

50%

Treatment B

AI Ranking

50%

Randomization Unit

Configure:

- Customer
- Session
- Transaction
- Visit
- Account

TICVAI should avoid assigning the same customer inconsistently where the experiment design requires persistent assignment.

Audience

Example:

Family Visitors

Venue:

Dubai Theme Park

Channel:

B2C

Experiment Duration

- Start Date
- End Date
- Minimum Sample
- Maximum Duration

Primary Metric

Fast Pass Attach Rate

Secondary Metrics

- Revenue
- AOV
- Conversion
- Customer Rejection
- Checkout Completion

Guardrail Metrics

Examples:

- Checkout Abandonment
- Refund Rate
- Recommendation Dismissal
- Transaction Time
- Service Complaints where available

Exclusions

Exclude:

- Employees
- Test Accounts
- Invalid Sessions
- Internal Traffic
- Other configured populations

Actions

Create Experiment | Validate Design | Start Experiment | Save Draft

Screen 6 — Experiment Results & Incrementality Analysis

Purpose

Measure whether a recommendation actually created incremental improvement.

Experiment Header

Fast Pass AI Ranking Test

Status:

Completed

Sample:

120,000 Sessions

Results

Metric	Control	Treatment	Difference
Attach Rate	8.2%	10.6%	+2.4 pp
Conversion	71.8%	72.1%	+0.3 pp
AOV	AED 438	AED 462	+AED 24
Revenue/Session	AED 314	AED 333	+AED 19

Statistical Result

Where TICVAI implements valid statistical analysis, show:

- Sample Size
- Effect Size
- Confidence Interval
- Statistical Significance / Probability
- Experiment Duration

Avoid reducing experimentation to only:

Winner ✓

The administrator needs enough evidence to understand whether the result is meaningful.

Incremental Revenue

Example:

Treatment generated:

AED 19 additional revenue per eligible session versus control.

Eligible treatment population:

60,000

Estimated incremental revenue:

AED 1.14M

subject to the experiment methodology.

Guardrail Results

Example:

Checkout Abandonment

Control: 8.1%

Treatment: 8.4%

Change:

+0.3 pp

Status:

Within Allowed Threshold

Segment Analysis

Analyze result by:

- Customer Type
- Audience
- Channel
- Venue
- Journey Stage
- Device
- Membership
- New vs Returning

But warn against drawing conclusions from very small subgroups.

Decision

Options:

Promote Treatment

Continue Test

Stop Test

Reject Treatment

Run Follow-Up

Actions

View Methodology | Compare Segments | Promote Winner | Export Results

Screen 7 — Strategy, Model & Decision Mode Comparison

Purpose

Compare recommendation approaches to understand which intelligence strategy performs best.

Compare

Examples:

AI Ranked

vs

Rule-Based

vs

Hybrid

vs

Business Priority

Performance Table

Mode	Acceptance	Conversion	AOV Uplift	Incremental Revenue	Confidence	Latency
AI	12.8%	10.4%	+8.2%	AED 620K	86%	142ms
Hybrid	13.4%	10.9%	+8.8%	AED 690K	89%	126ms
Rules	9.1%	8.2%	+4.1%	AED 310K	—	18ms
Priority	8.7%	7.9%	+3.8%	AED 280K	—	12ms

Model Comparison

Where Board 3 uses different models/providers:

- Model Version
- Recommendation Quality
- Conversion
- Revenue
- Latency
- Failure Rate
- Cost per Decision where available

Important

The highest revenue model should not automatically be considered best.

TICVAI should allow evaluation across:

Commercial Impact

-

Customer Experience

-

Latency

-

Reliability

-

Cost

-

Business Constraints

Pareto View

Visualize:

Performance vs Cost

or:

Conversion vs Latency

Actions

Compare Modes | Compare Models | Open Experiment | Recommend Candidate for Promotion

Screen 8 — Recommendation Quality & Optimization Opportunity Center

Purpose

Automatically identify where recommendation strategies appear to be underperforming and generate controlled optimization opportunities.

This is where AI can become useful for administrators.

Opportunity Categories

- Low Conversion
- Low Acceptance
- High Suppression
- Low Confidence
- Excessive Fallback
- Poor Candidate Diversity
- Placement Underperformance
- Audience Mismatch
- Channel Underperformance
- High Repetition
- Low Candidate Availability
- High Decision Latency
- Recommendation Fatigue

Example Opportunity

Family Meal — Checkout

Observed:

High impressions

Low acceptance

AI analysis:

Customers interact more strongly with this recommendation at Basket than at Checkout.

Suggested action:

Test moving the primary Family Meal placement from Checkout to Basket.

Another Example

Annual Membership

Observed:

Strong conversion among returning visitors

Weak conversion among first-time visitors

Recommendation:

Consider testing different strategy priority by audience.

Optimization Card

Each opportunity should show:

- Finding
- Evidence
- Affected Strategy
- Affected Audience
- Potential Action
- Confidence
- Expected Area of Impact
- Risk

Decision

Administrator can:

Create Experiment

Prepare Configuration Change

Dismiss

Defer

Ask AI Why

Important

AI should not silently change recommendation strategies because it detected underperformance.

Optimization remains governed.

Actions

Review Opportunity | Create Experiment | Prepare Change | Dismiss

Screen 9 — Recommendation Learning & Feedback Loop Governance

Purpose

Control how performance outcomes are fed back into AI/model decisioning.

This is critical. TICVAI should not simply “let the AI learn automatically” from every click.

Feedback Events

Potential learning signals:

- Impression
- Click
- Acceptance
- Rejection
- Purchase
- Removal
- Refund
- Upgrade
- Renewal
- Completed Action
- Recommendation Suppression
- No Response

Signal Meaning

Administrators must understand that:

Purchase

is a strong positive signal.

But:

No Click

is not necessarily a negative signal.

And:

Refund

may require different treatment depending on cause.

Feedback Configuration

Event	Learning Use	Strength	Delay	Status
Purchase	Positive	Strong	Immediate	Enabled

Event	Learning Use	Strength	Delay	Status
Accepted	Positive	Medium	Immediate	Enabled
Explicit Reject	Negative	Medium	Immediate	Enabled
No Click	Neutral	—	—	Enabled
Refund	Review	Contextual	Delayed	Enabled

Learning Modes

Support:

Offline Learning

Performance data is periodically used to train/recalibrate models.

Controlled Online Learning

Approved models adapt within defined boundaries.

No Learning

Results used only for reporting.

Model Promotion

Recommended lifecycle:

Production Model

↓

Collect Outcomes

↓

Train Candidate Model

↓

Offline Validation

↓

Experiment / Champion-Challenger

↓

Approval

↓

Production Promotion

Champion / Challenger

Example:

Champion

Recommendation Model v3.2

Challenger

Recommendation Model v3.3

Traffic:

90% / 10%

Drift Indicators

Monitor:

- Feature Drift
- Audience Drift
- Conversion Drift
- Confidence Drift
- Candidate Distribution Change

Detailed platform-level model governance can later sit in the AI Platform/Governance module.

Actions

[Configure Feedback](#) | [View Learning Policy](#) | [Compare Champion/Challenger](#) | [View Drift](#)

Screen 10 — Optimization Plan, Approval & Performance Simulation

Purpose

Bring Board 5 together by converting validated insights into controlled improvement actions.

This is the final screen of the entire Recommendation & Personalization module.

Optimization Summary

Example:

Current Strategy

Family Visit Recommendation

Performance Status

Healthy with Opportunities

Identified Opportunities

1. Family Meal Placement

Move:

Checkout → Basket

Evidence:

+18% engagement observed at Basket.

Recommended action:

Run controlled placement experiment

2. Annual Membership

Increase recommendation priority for:

Returning Non-Members

Evidence:
Higher historical conversion in this audience.
Recommended action:

Create audience-specific experiment

3. Fast Pass

Current performance:
Strong
Recommendation:

No Change

This is important—AI should also be able to recommend doing nothing.

Optimization Plan

Change	Source	Risk	Validation	Approval
Test Meal Placement	AI Opportunity	Low	Ready	Required
Membership Audience Test	AI Opportunity	Medium	Ready	Required
Fast Pass	No Change	—	—	—

Expected Impact

Where supported by evidence:

- Expected Conversion Range
- Expected Revenue Range
- Affected Audience
- Affected Channels
- Operational Risk

These should be represented as **estimates**, not guarantees.

Simulation

Before implementing:

Current Configuration

vs

Proposed Configuration

Show expected differences.

Approval

Optimization changes should route through applicable TICVAI governance.
For example:

Placement Test

→ Recommendation Owner

Commercial Priority Change

→ Commercial Owner

Model Promotion

→ AI / Platform Governance

Apply Change

Do not directly manipulate the underlying Board 1–4 configuration.

Instead:

Approved Optimization

↓

Create controlled change request

↓

Send to owning configuration

Examples:

Strategy Change → Board 1

Context/Audience Change → Board 2

Decision/Ranking Change → Board 3

Placement/Delivery Change → Board 4

Closed Loop

After activation:

New Version

↓

Measure

↓

Compare

↓

Learn

↓

Optimize again

Actions

Create Optimization Plan | Run Simulation | Submit for Approval | Create Experiment | View History

Critical Performance Architecture

Soft Labs should preserve the IDs generated throughout Boards 1–4.

Without this, Board 5 cannot perform reliable analytics.

The lineage should be:

Strategy ID

↓

Context ID

↓

Decision ID

↓

Delivery ID

↓

Interaction ID

↓

Transaction / Action ID

↓

Attribution

↓

Experiment Assignment

↓

Performance Outcome

For example:

STR-1008

↓

CTX-88210

↓

DEC-49102

↓

DLV-76211

↓

INT-90018

↓

ORD-28210

This provides complete recommendation lineage.

Recommendation Performance Event Model

I recommend a standardized event model such as:

RecommendationEvent

|

├— Event ID

├— Event Type

|— Timestamp
|
|— Strategy ID
|— Context ID
|— Decision ID
|— Delivery ID
|
|— Customer / Session Reference
|— Venue
|— Channel
|— Journey Stage
|— Placement
|
|— Recommended Product / Action
|— Rank
|— Decision Mode
|— Model Version
|
|— Interaction
|— Transaction / Action Reference
|— Revenue
|— Currency
|
|— Experiment Assignment

That common event model will make Board 5 much more powerful.

Attribution vs Incrementality

This should be explicitly explained to Soft Labs because it is easy to design incorrectly.

Attribution asks:

Which recommendation should receive credit for this purchase?

Example:

Customer clicks Fast Pass recommendation and purchases it.

Attributed Revenue = AED 120

Incrementality asks:

Would the customer have purchased Fast Pass anyway without the recommendation?

This requires stronger evidence, usually from experiments/control groups or appropriate causal methodology.

Suppose:

Treatment conversion = **12%**

Control conversion = **9%**

Incremental effect $\approx$ **3 percentage points**

Therefore not all revenue from the 12% treatment conversion should automatically be claimed as AI-generated incremental revenue.

TICVAI should display these as separate metrics.

Recommendation Revenue Definitions

I recommend clearly defining at least:

Attributed Revenue

Revenue connected to recommendation interactions according to the selected attribution model.

Incremental Revenue

Estimated/measured additional revenue generated because of the recommendation intervention.

Assisted Revenue

Revenue where recommendation interaction was part of the journey but did not receive primary attribution.

This avoids inflated AI performance claims.

Experiment Integrity

The system should prevent common mistakes such as:

- Customer switching between Control and Treatment unexpectedly
- Overlapping experiments creating contamination
- Stopping tests only when results look favorable
- Tiny sample sizes being declared winners
- Comparing periods with fundamentally different operating conditions without warning

Experiment Collision Detection

Example:

Experiment A

Fast Pass Ranking

Audience:

Family Visitors

Experiment B

Checkout Placement

Audience:

All Visitors

TICVAI detects overlap and displays:

Potential Experiment Interaction

The administrator can:

- Allow
 - Make Mutually Exclusive
 - Change Audience
 - Sequence Experiments
-

AI Optimization Safety

The optimization loop should be:

Observe

→ **Analyze**

→ **Recommend**

→ **Experiment**

→ **Validate**

→ **Approve**

→ **Deploy**

→ **Measure**

not:

Observe → AI automatically changes production.

Later, TICVAI may support controlled low-risk auto-optimization, but that should be governed by the broader **AI Governance** framework.

Feedback Quality

Board 5 should not teach the model that every commercial outcome is desirable.

For example:

AI recommends an expensive upgrade.

Customer purchases it.

Then immediately requests a refund.

A simplistic model sees:

Purchase = Success

A better system can eventually understand:

Purchase + Immediate Refund = potentially poor outcome

Similarly, a recommendation increasing revenue but significantly increasing checkout abandonment may not be optimal.

Therefore optimization should support **multi-objective outcomes**, not only revenue.

Multi-Objective Optimization

Potential objectives include:

- Revenue
- Conversion
- Margin
- Attach Rate
- Membership Conversion
- Customer Relevance
- Experience Utilization
- Retention
- Checkout Completion
- Recommendation Acceptance
- Operational Utilization

Board 1 defines the strategic objective.

Board 5 measures whether it was achieved.

Important Boundary — Reporting & BI

Board 5 provides **specialized recommendation intelligence analytics**.

Enterprise Reporting / Power BI can consume these metrics for broader dashboards.

For example:

Board 5

Detailed Fast Pass recommendation experiment.

Enterprise BI

Total venue revenue including recommendation contribution.

Do not duplicate the entire enterprise reporting platform here.

Important Boundary — AI Governance

Board 5 may identify:

Model v3.3 outperforms v3.2.

But promotion into production may require:

AI Governance Approval

depending on TICVAI policy.

This is particularly important when we later design the separate AI Governance area.

Important Boundary — AI Platform

Board 5 should measure:

- Model Version

- Performance
- Latency
- Failure Rate
- Decision Cost where available

But:

- Provider Credentials
- Model Deployment
- Model Endpoint
- Provider Routing
- Token/compute management
- Model infrastructure

belong to the future **AI Platform, Models & Providers** area.

Board 5 — Final Screen Map

Recommendation Performance, Experimentation & Optimization

1. Recommendation Performance Command Center
2. Recommendation Funnel & Engagement Analytics
3. Revenue, Conversion & Commercial Impact Analytics
4. Recommendation Attribution & Conversion Mapping
5. Experiment & Control Group Designer
6. Experiment Results & Incrementality Analysis
7. Strategy, Model & Decision Mode Comparison
8. Recommendation Quality & Optimization Opportunity Center
9. Recommendation Learning & Feedback Loop Governance
10. Optimization Plan, Approval & Performance Simulation

Recommendation & Personalization Module — COMPLETE

We now have a complete 5-board / 50-screen architecture:

Board 1 — Recommendation Strategy & Configuration

Business objectives, use cases, candidates, eligibility, relationships, constraints, conflicts and fallbacks.

Board 2 — Customer, Context & Audience Intelligence

Identity, behavior, session, basket, party, visit, audience, real-time context and permitted data usage.

Board 3 — Recommendation Decisioning, Ranking & AI Intelligence

Candidate funnel, AI scoring, ranking, next-best-action, diversity, confidence and explainability.

Board 4 — Personalized Offers & Omnichannel Journey Delivery

Journey, placements, triggers, presentation, frequency, omnichannel orchestration, validation and response capture.

Board 5 — Recommendation Performance, Experimentation & Optimization

Performance, attribution, incrementality, experimentation, optimization and controlled learning.

Together, these boards give TICVAI a full closed-loop recommendation architecture:

Understand → Decide → Deliver → Measure → Learn → Improve.